UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II



SCHOOL OF ARCHITECTURE

Master's Degree in DESIGN FOR THE BUILT ENVIRONMENT

Department of Architecture

Predictive Mood Tracking App for Bipolar Disorder Patients

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ACADEMIC YEAR 2024/2025

Abstract

Bipolar disorder (BD) is a complex psychiatric condition characterized by significant mood swings, including episodes of mania, hypomania, and depression. The effective management of BD presents ongoing challenges, particularly in terms of early intervention, mood stabilization, and medication adherence. Traditional methods for monitoring and managing BD rely on infrequent clinical visits and manual mood tracking, often leading to delayed interventions and poor treatment outcomes. This thesis presents the development of a Predictive Mood Tracking App designed to enhance the management of BD by integrating AI-driven mood prediction, personalized feedback, and collaborative care features.

The primary objective of this research was to design and develop an innovative mobile application that utilizes AI to predict mood fluctuations based on self-reported data, such as emotional states, sleep patterns, and daily activities. The app aims to provide proactive interventions, such as personalized motivational messages and activity suggestions, helping users regulate their emotional well-being and manage mood shifts more effectively. Furthermore, the app allows for collaborative care, enabling patients to share their mood data with healthcare providers and trusted support persons to improve treatment adherence and communication.

The research methodology adopted a user-centered design approach, incorporating literature review, clinical insights, and analysis of existing solutions to identify key user needs. The design process was informed by best practices in digital health and AI-based mood prediction models, ensuring that the app meets both the clinical requirements and emotional support needs of individuals with BD. The app was prototyped using Figma for UI/UX design, with a focus on usability, accessibility, and emotional engagement.

The findings of this research demonstrate that the AI-powered mood prediction system can significantly enhance early intervention by forecasting manic and depressive episodes based on user-reported data. The app also provides users with

personalized support through motivational messages and tailored suggestions for managing emotional fluctuations. However, future development should focus on integrating real-time user feedback, biometric data, and advanced AI algorithms to further improve the predictive accuracy and personalization of interventions.

This thesis contributes to the field of digital mental health by demonstrating the potential of AI to enhance mood tracking and provide personalized emotional support for individuals with bipolar disorder. The development of such AI-driven applications has the potential to revolutionize the way bipolar disorder is managed, offering users a comprehensive tool for proactive mental health care and improving long-term well-being.

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Chapter 1

Introduction

1.1 Background and Importance of Bipolar Disorder

Bipolar disorder (BD) is a chronic psychiatric illness marked by alternating episodes of elevated mood (mania or hypomania) and depression, affecting an estimated 45 million people worldwide and ranking among the leading causes of disability-adjusted life years lost [1]. The DSM-5 defines a manic episode as a distinct period of abnormally elevated, expansive, or irritable mood and increased goal-directed activity or energy, lasting at least one week (or any duration if hospitalization is required), accompanied by at least three of the following symptoms: inflated self-esteem or grandiosity; decreased need for sleep; more talkative than usual; flight of ideas; distractibility; increased goal-directed activity or psychomotor agitation; and engagement in high-risk activities [2]. A hypomanic episode meets identical symptom criteria but persists for at least four days without causing marked functional impairment or necessitating hospitalization. A major depressive episode requires at least five of nine symptoms including depressed mood or anhedonia over a 2-week period, such as significant changes in appetite or sleep, psychomotor changes, fatigue, feelings of worthlessness, impaired concentration, or recurrent thoughts of death [2].

BD is subclassified into three primary subtypes:

- **Bipolar I Disorder:** at least one manic episode, with or without major depressive episodes.
- **Bipolar II Disorder:** at least one hypomanic episode and one major depressive episode, with no history of full manic episodes.
- **Cyclothymic Disorder:** at least two years of numerous periods of subthreshold hypomanic and depressive symptoms that do not meet full episode criteria [2].

Accurate and timely diagnosis is often hindered by symptomatic overlap with major depressive disorder and other mood disorders, leading to delayed or incorrect treatment and poorer long-term outcomes [3].

Neurobiological Underpinnings

Converging evidence supports a multifactorial etiology in BD. First-degree relatives of BD patients face a ten-fold increased risk, and genome wide association studies implicate ion-channel and synaptic genes most notably *CACNA1C* in susceptibility [4]. Dysregulation of key neurotransmitter systems (serotonin, norepinephrine, dopamine) and imbalances in glutamate/GABA transmission have been documented across mood phases [5]. Structural and functional neuroimaging consistently reveal reduced volumes in the volumes in the prefronhippocampus,campus, amygdala hyperactivitaltered frontolimbic white mattertter tracts [6]. Finally, hyperactivity of the hypothalamic-pituitary-adrenal axis and variants of the circadian rhythm gene contribute to mood instability and the episodic course of BD [7].

Given these diagnostic and biological complexities, there is a pressing need for innovative digital solutions capable of continuous monitoring, early prediction, and personalized intervention to improve the management and prognosis of BD.

1.2 The Role of Digital Health in Mental Health Management

Recent advancements in digital health technologies have introduced continuous, real-time monitoring capabilities that address the episodic and limited nature of traditional bipolar disorder (BD) assessments [8, 9]. Various smartphone applications currently available differ significantly in their approaches and overall effectiveness. For instance, *Bipolar Mood Tracker* limits mood tracking to binary yes/no entries, while *Mood Journal* employs a scale-based rating system but lacks the capability to integrate physiological data (see Chapter 2, Table 2.2). Additionally, applications such as *eMoods* capture mood fluctuations along with limited symptom details but do not include medication reminders or sleep tracking features. Even more comprehensive tools, such as *PolarUs* which offers sleep tracking and advanced data visualization still fall short by not incorporating predictive AI models capable of proactively forecasting mood episodes and providing personalized interventions (see Chapter 2, Table 2.2).

These identified limitations particularly the absence of multimodal data integration and predictive analytical capabilities underscore the critical need for more advanced, holistic, and user-centered digital health solutions. The subsequent chapter reviews technical architectures and design principles that specifically aim to bridge these gaps, thus guiding the development of the prototype introduced in this thesis.

1.3 Research Problem and Objectives

Although digital health solutions have significantly advanced, existing mood tracking applications for bipolar disorder often rely primarily on simple, user-logged mood entries without utilizing the predictive power of advanced AI models. Current solutions typically lack real-time analytical capabilities necessary to forecast mood episodes accurately and provide users with personalized, proactive interventions tailored to their emotional states.

To address these identified gaps, this research seeks to achieve the following objectives:

- Evaluate the limitations of current bipolar disorder tracking applications, particularly concerning their data collection methodologies and predictive capabilities.
- Investigate how AI-driven models and digital health tools can enhance mood prediction through comprehensive analysis of user-reported data, including emotional states, sleep patterns, medication adherence, and daily activities.
- Propose a robust conceptual framework and user-centered design for integrating AI-driven mood forecasting into a mobile health application, providing proactive mental health insights and personalized interventions to users.

1.4 Structure of the Thesis

This thesis is structured into five chapters, providing a comprehensive exploration of digital health technologies and their application to bipolar disorder management:

- **Chapter 2: Literature Review** – Provides an in-depth analysis of existing research on bipolar disorder classification, current clinical treatment approaches, digital mental health interventions, and the latest developments in AI-driven mood prediction models.
- **Chapter 3: Methodology** – Outlines the research design, data collection methods, AI predictive modeling concepts, and the system architecture underpinning the proposed predictive mood tracking application.
- **Chapter 4: Discussion and Analysis** – Examines existing bipolar disorder tracking applications, identifying strengths and weaknesses, and critically explores the feasibility and benefits of integrating predictive AI-based mood forecasting within a digital health framework.

- **Chapter 5: Results and Conclusion** – Summarizes the key findings, highlights the contributions of this research to the fields of digital health and mental health management, and proposes future research directions and potential improvements to further enhance the app's capabilities and user experience.

This thesis contributes to advancing digital mental health care by demonstrating how AI can significantly improve the accuracy and efficacy of mood prediction and early intervention in bipolar disorder management. By developing a comprehensive, predictive, and user-centered digital health solution, this research aims to improve the lives of individuals with bipolar disorder, promoting better emotional regulation, improved adherence to treatment plans, and enhanced overall mental well-being.

1.5 Study Scope and Boundaries

While this thesis develops a conceptual framework and prototype for AI-driven mood prediction in bipolar disorder, it deliberately excludes several areas of applied research and development. No new human subjects research, patient trials, or real-world clinical studies are performed to validate model efficacy, safety, or diagnostic accuracy. The work does not extend to production-level implementation of a fully operational application (including back-end integration, user-interface hardening, data-security infrastructure, or maintenance workflows). Longitudinal follow-up studies, cost-effectiveness analyses, and health economics modeling lie outside its remit, as do the design, testing, and evaluation of end-user training materials (e.g. user manuals, tutorial videos) and deployment logistics in clinical or home settings. Finally, this thesis does not conduct qualitative stakeholder research to assess lived experience, perceived barriers, or acceptance factors among patients, clinicians, or caregivers. By clarifying these boundaries up front, the reader is able to focus on the technical architectures, simulated validations, and ethical/regulatory foundations that constitute the core contributions of this research.

Chapter 2

Literature Review

2.1 Introduction to Bipolar Disorder

2.1.1 Definition and Classification

Bipolar disorder (BD) is a chronic mental illness characterized by manic, hypomanic, and depressive episodes that significantly impact daily functioning. It is classified into Bipolar I Disorder (characterized by full manic episodes), Bipolar II Disorder (hypomania and major depressive episodes), and Cyclothymic Disorder, which involves less severe but persistent mood instability [10]. Recent research highlights that BD shares overlapping symptoms with major depressive disorder (MDD), leading to frequent misdiagnoses and delayed treatment initiation [11].

2.1.2 Symptoms and Mood Cycles

The disorder is marked by alternating mood cycles, including manic, depressive, and mixed states. Manic episodes involve euphoria, impulsivity, reduced need for sleep, and increased goal-directed activity, whereas depressive episodes manifest as persistent sadness, social withdrawal, and suicidal ideation [10]. Some patients experience rapid cycling, where four or more mood episodes occur within a year, complicating treatment response [11]. Studies suggest that early intervention and predictive AI models can help manage these fluctuations by forecasting mood episodes [8]. Understanding these mood cycles and rapid shifts is critical for

developing predictive AI tools that can anticipate and manage these episodes proactively, aligning directly with the core objectives of this thesis.

Symptoms of a Manic Episode	**Symptoms of a Major Depressive Episode**
Feeling very up, high, elated, extremely irritable, or touchy	Feeling very down or sad, or anxious
Feeling jumpy or wired, or being more active than usual	Feeling slowed down or restless
Racing thoughts	Trouble concentrating or making decisions
Decreased need for sleep	Trouble falling asleep, waking up too early, or sleeping too much
Talking fast about a lot of different things ("flight of ideas")	Talking very slowly, feeling unable to find anything to say, or forgetting a lot
Excessive appetite for food, drinking, sex, or other pleasurable activities	Lack of interest in almost all activities
Feeling able to do many things at once without getting tired	Unable to do even simple things
Feeling unusually important, talented, or powerful	Feeling hopeless or worthless, or thinking about death or suicide
Difficulty maintaining work responsibilities, social activities, or relationships	

Table 2.1: Comparison of symptoms in manic versus major depressive episodes [10, 11]

2.1.3 Causes and Risk Factors

BD is influenced by genetic predisposition, neurobiological factors, and environmental triggers. Genetic studies estimate an 80% heritability rate, linking BD to neurotransmitter dysregulation and neurostructural abnormalities in the prefrontal cortex, amygdala, and hippocampus [10]. Environmental stressors such as sleep disturbances, childhood trauma, and substance abuse can trigger mood episodes [11]. Advancements in AI-based neuroimaging analysis have enhanced our ability to detect BD related brain patterns, opening pathways for more precise diagnosis and treatment personalization [8].

Sleep plays a crucial role in mood stability, with research indicating that irregular sleep patterns can act as a precursor to mood episodes [12]. AI-powered sleep tracking and predictive analytics are emerging as valuable tools for detecting early warning signs of mood shifts based on sleep irregularities.

2.2 Current Management and Tracking of Bipolar Disorder

2.2.1 Traditional Treatment Approaches (Medication, Therapy)

BD management traditionally involves mood stabilizers (e.g., lithium, valproate), atypical antipsychotics, and antidepressants [10]. Treatment adherence is a persistent challenge, with up to 40% of BD patients discontinuing their medication due to side effects or lack of perceived effectiveness [11]. Cognitive Behavioral Therapy (CBT) and psychoeducation have been shown to improve long-term outcomes, especially when integrated with pharmacological treatments [10].

2.2.2 The Rise of Digital Health Solutions

The adoption of digital mental health technologies has surged, particularly after the COVID-19 pandemic, which accelerated remote mental health care [13]. AI

driven symptom trackers and teletherapy platforms have significantly improved access to psychiatric support. Unlike traditional mental health assessments, which rely on infrequent clinical visits, digital health solutions enable continuous monitoring and early intervention [14].

While these solutions improve accessibility, key challenges remain, including data privacy, user adherence, and ethical concerns in AI-based mental health applications [15]. Notably, AI-powered medication adherence tools have been explored as a means to enhance treatment compliance [16]. These tools can track whether patients are taking their prescribed medications, reducing the likelihood of relapse due to nonadherence.

Beyond technological functionality, recent research emphasizes that the success of digital health applications also depends heavily on their cultural sensitivity and user-centered design. The importance of culturally sensitive and user-centered design in digital health interventions is increasingly recognized. Asif, Malik, and Khan [17] developed the culturally adapted PsychoEducation (CaPE) mobile application to improve adherence to treatment among patients with bipolar disorder in Pakistan. Their study underscores the necessity of incorporating stakeholder feedback, cultural relevance, and user-friendly design principles when developing mental health applications, ensuring accessibility and emotional resonance among diverse user populations [17].

In addition to cultural adaptation, the success of digital health applications relies on applying broader user-centered design principles to improve usability, engagement, and clinical effectiveness. Grill, Smith, and Jones [18] emphasize that the incorporation of personalized features such as tailored feedback, mood-adaptive suggestions, and individual goal setting enhances user engagement and intervention efficacy. Additionally, Fleming et al. [19] highlight the role of supportive community features, noting that opportunities for social interaction within digital platforms can significantly strengthen users' emotional resilience and well-being. Accessibility and usability also represent key priorities; Bakker et al. [20] argue that intuitive navigation, clear interface design, and inclusive accessibility standards enable broader reach among diverse user populations.

Finally, Hollis et al. [21] stress the importance of integrating evidence-based therapeutic techniques into the app structure to enhance its clinical effectiveness. Collectively, these design principles form the foundation for creating effective, engaging, and impactful digital interventions for individuals managing bipolar disorder. [18, 19, 20, 21]

These developments establish the groundwork upon which this thesis aims to build by creating an AI-driven app that integrates real-time mood prediction with personalized digital interventions.

2.2.3 The Role of Family/Loved Ones in BD Management

Social support plays a critical role in BD management. Family members and loved ones not only provide emotional support, but also act as early warning systems by recognizing mood changes before they escalate [22]. Digital tools that allow caregivers to be notified of significant mood changes could further enhance patient support networks.

2.3 Comparison of Existing Bipolar Tracking Applications

2.3.1 Overview of Market-Available Apps

Various mobile applications help people with bipolar disorder (BD) track their moods, symptoms, and related factors. These apps aim to improve self-monitoring and help healthcare professionals manage BD. Below is an analysis of five commonly used bipolar tracking applications.



Figure 2.1: Feature Comparison of Existing Bipolar Disorder Tracking Applications

This chart summarizes the presence or absence of key features across five widely used bipolar tracking applications, highlighting gaps in AI integration, early warning alerts, and emotional support tools.

- **Bipolar Mood Tracker** – A simple, user-friendly mood tracking app that allows manual data entry but lacks advanced insights.
- **Bipolar UK Mood Tracker** – A scale-based tracking system that enables users to log daily emotions and review trends over time.
- **eMoods** – Tracks mood fluctuations, highs/lows, and symptoms but has limited AI integration.
- **PolarUs** – A comprehensive application focusing on various life aspects, including sleep, lifestyle, relationships, and cognitive well-being.
- **Bipolar Mood Tracker BPD Diary** – A detailed tracking tool with a scoring system, historical records, and multiple symptom tracking.

2.3.2 Strengths and Weaknesses of Bipolar Tracking Apps

Bipolar Mood Tracker APP

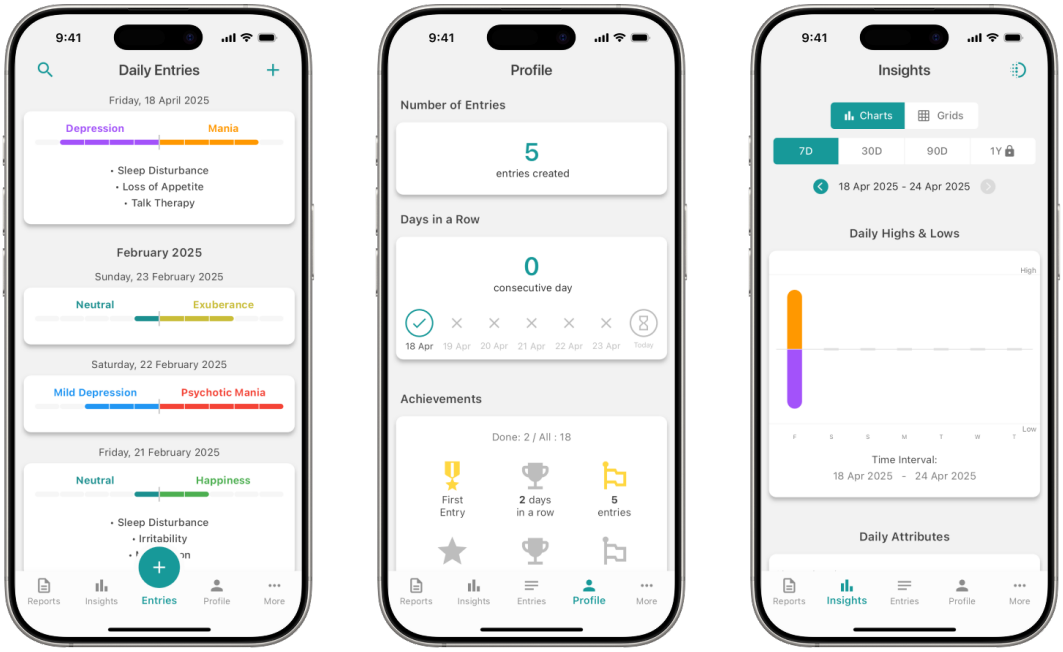
This application is designed for quick and simple mood tracking. Users can enter their mood manually, generate reports, and track attributes such as sleep disturbances, medication, and therapy sessions. However, the app only offers Yes/No responses for attributes, limiting the depth of analysis. It provides privacy features such as Face ID and Touch ID lock, along with gamification elements like achievements and daily reminders.

Strengths:

- Simple and user-friendly interface.
- Secure with Face ID/Touch ID protection.
- Generates quick PDF/CSV reports for medical professionals.

Weaknesses:

- Lacks AI-driven mood analysis.
- No sleep-tracking or medication reminder features.
- Binary Yes/No tracking system is too limited.



(a) Home screen (b) Mood-entry interface (c) Report export preview

Figure 2.2: Bipolar Mood Tracker App interface overview

Bipolar UK Mood Tracker App

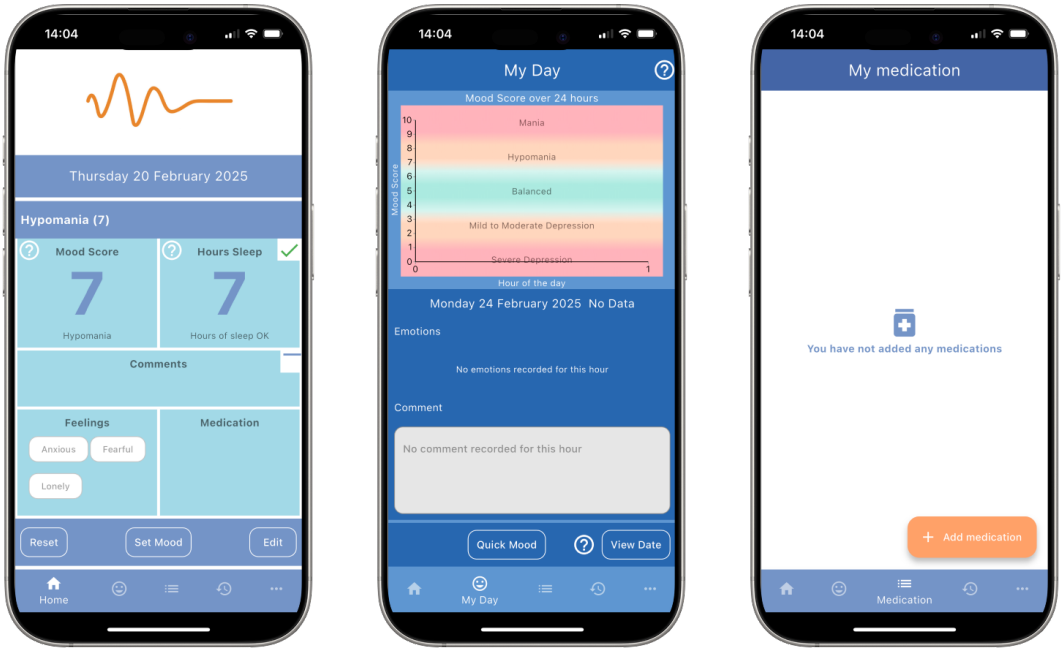
This app uses a scale-based rating system for mood tracking, allowing users to pick from a range of emotions rather than just Yes/No responses. It includes medication tracking with reminders, making it useful for managing treatment. It also provides weekly and monthly insights for reviewing mood trends.

Strengths:

- Uses a rating scale for mood tracking, improving accuracy.
- Allows users to track sleep hours and monitor sleep trends.
- Includes medication reminders.

Weaknesses:

- Limited customization options for symptom tracking.
- No AI-based mood prediction or automated insights.
- No in-app community support.



(a) Home screen (b) My Day view (c) Medication screen

Figure 2.3: Bipolar UK Mood Tracker App interface overview

eMoods Bipolar Mood Tracker APP

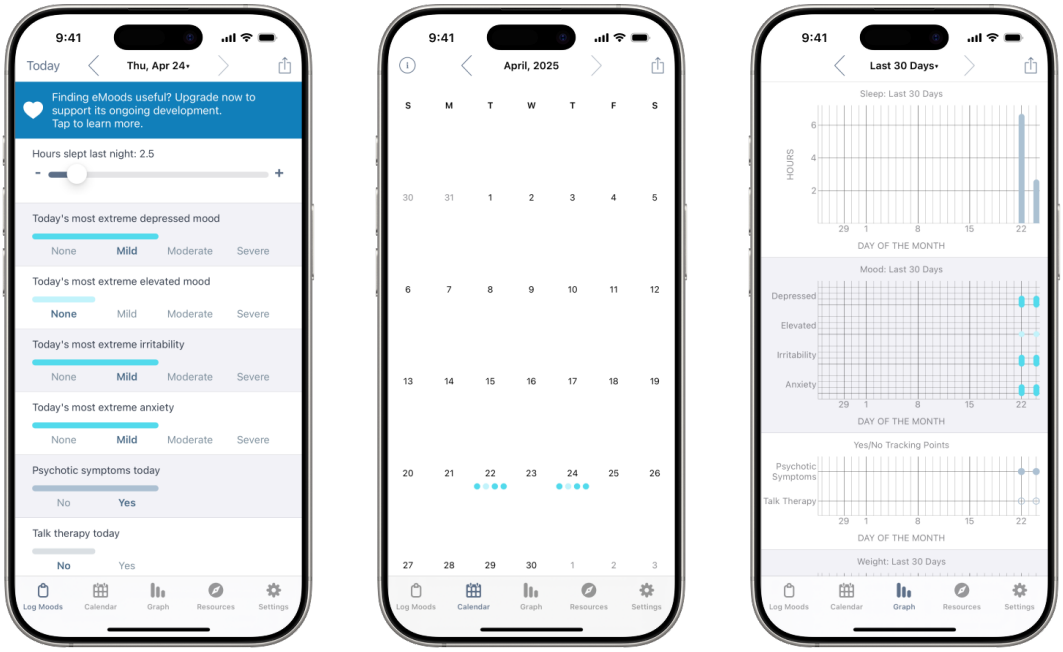
eMoods is a privacy-focused app that does not store data in the cloud. It allows users to track mood swings, sleep hours, and symptoms such as anxiety, irritability, and depression. The app generates color-coded symptom charts for easy analysis but lacks automated AI insights.

Strengths:

- Strong privacy protection—data is stored locally.
- Professional reports in PDF format.
- Color-coded symptoms for easy trend analysis.

Weaknesses:

- No built-in medication reminders.
- No AI-driven mood predictions.
- Lacks detailed sleep tracking.



(a) Mood-logging screen (b) Calendar overview (c) 30-day trends graph

Figure 2.4: eMoods Bipolar Mood Tracker APP interface overview

PolarUs App

PolarUs goes beyond mood tracking by monitoring various aspects of quality of life and behavioral patterns. Unlike other apps, it provides expert-backed strategies and educational resources. However, the app may feel overwhelming due to its many features.

Strengths:

- Monitors multiple aspects of life beyond mood tracking.
- Provides expert advice and scientifically-backed strategies.
- Includes sleep tracking and lifestyle monitoring.

Weaknesses:

- May feel complex for users looking for simple mood tracking.
- No AI-driven insights or mood predictions.
- Lacks peer support or interactive community features.



(a) Resources screen (b) Home dashboard (c) My Strategies library

Figure 2.5: PolarUs App interface overview

Mood Journal BPD App

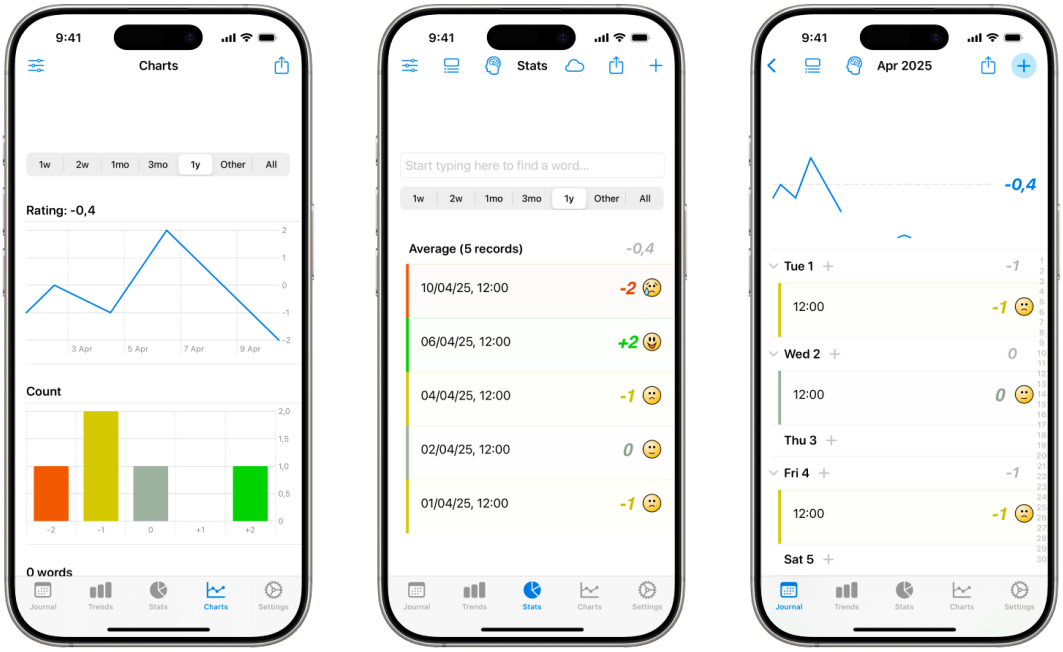
This app is designed for individuals with bipolar disorder, depression, and anxiety. It offers a detailed symptom-tracking system with a scoring method. Users can record their daily mood swings, symptoms, and activities such as alcohol use, therapy attendance, and missed medications.

Strengths:

- Uses a rating system for symptom tracking.
- Provides historical insights over different timeframes.
- Helps users identify triggers and patterns.

Weaknesses:

- No sleep-tracking functionality.
- No calendar function for reviewing past moods.
- Lacks gamification features.



(a) Charts overview (b) Stats list (c) Journal timeline

Figure 2.6: Mood Journal BPD App interface overview

Summary of App Comparisons

The reviewed apps each offer valuable features but suffer from key limitations that reduce long-term engagement. Simpler tools such as **Bipolar Mood Tracker** are easy to use but fail to provide personalized insights, while more comprehensive platforms like **PolarUs** are praised for depth but criticized for complexity. Most notably, none of the reviewed apps currently integrate AI-based predictions, real-time alerts, or multimodal data fusion features central to this thesis. Additionally, user feedback often points to usability issues, limited flexibility, and lack of emotional resonance as common shortcomings. These insights underscore the opportunity to design a more adaptive, predictive, and user-friendly application.

Table 2.2: Comparison of Market-Available Bipolar Disorder Tracking Apps and User Feedback

Feature	**Bipolar Mood Tracker**	**Bipolar UK Mood Tracker**	**eMoods**	**PolarUs**	**Mood Journal**
Mood Tracking Method	Yes/No options	Scale-based rating	Tracks highs/lows	Tracks multiple areas	Scale rating
Medication Tracking	No reminders	Reminders	None	None	None
Sleep Tracking	Not included	Sleep hours	Not included	Included	Not included
Data Privacy & Security	Face ID / Touch ID	No explicit features	Stored locally	Focus on privacy	No features
AI-Based Insights	None	None	None	None	None
Customizability	Yes/No only	Pick from list	Custom tracking	Personalized	Customizable symptoms
Data Visualization	Simple graphs	Mood/sleep graphs	PDF reports	Comprehensive	Bar charts
Community Support	No community	No community	No community	Educational resources	No community

(a) Feature Comparison

App Name	**General User Feedback**
Bipolar Mood Tracker	Easy to use, but lacks depth and personalization
Bipolar UK Mood Tracker	Positive for reminders; limited symptom flexibility
eMoods	Strong privacy; users note outdated interface
PolarUs	Comprehensive but overwhelming for some users
Mood Journal	Useful insights; missing features like sleep/calendar

(b) User Feedback Summary

2.3.3 Identified Gaps in Current Tracking Methods

Despite advancements, most BD tracking apps rely heavily on self-reported data, limiting predictive accuracy. Key gaps include:

- **Lack of AI-based mood prediction:** No reviewed apps use machine learning for forecasting mood episodes.
- **Limited integration with wearables:** Many apps do not sync with smart devices for real-time monitoring.
- **Absence of smart alerts:** No automatic notifications for high-risk mood fluctuations.

Future developments should integrate AI-driven insights, behavioral, biometric, and cognitive data to enhance mood forecasting.

In addition to the technical limitations identified, recent studies have emphasized the behavioral adaptations users employ to manage these gaps. Furthermore, Xu, Green, and Santos [23] explored user interactions within online bipolar disorder communities, revealing that many individuals independently adopt technology based strategies including mood-tracking apps and digital support platforms to manage their mental health. These findings highlight the growing reliance on digital tools among BD patients and reinforce the need for user-centered, integrated solutions like predictive mood-tracking applications [23].

Identifying these gaps underscores the necessity of the proposed predictive mood-tracking app, which integrates multimodal data analysis and advanced AI methodologies to deliver proactive mental health support.

2.4 AI-Based Mood Prediction in Bipolar Disorder

2.4.1 The Role of AI in Mental Health Monitoring

Artificial intelligence facilitates continuous mood tracking by analyzing user-reported symptoms, sleep patterns, and medication adherence. AI-driven models

can detect subtle changes in behavior, allowing for early warnings of depressive or manic episodes [11].Predictive analytics, which uses historical data to forecast mood fluctuations, is increasingly being integrated into mobile applications [24].

Recent advancements in artificial intelligence (AI) for mood prediction in bipolar disorder have increasingly prioritized the use of self-reported mood and behavior data, reflecting a shift toward more accessible and user-centered digital health solutions. Kannan, Smith, and Chen [25] demonstrate that the integration of deep learning and machine learning techniques has shown significant promise in enhancing diagnostic accuracy and treatment outcomes for bipolar disorder patients. These technologies focus on analyzing structured datasets collected directly from user-reported behavioral assessments, such as mood entries, sleep patterns, medication adherence, and daily activity logs. Rather than relying on invasive biometric monitoring or wearable devices, contemporary AI approaches emphasize the utility of longitudinal self-reported data. Predictive models are trained to detect subtle changes in user-reported mood states and behavioral patterns over time, enabling early detection of depressive, manic, or hypomanic episodes. This strategy aligns with the goals of personalized medicine by offering proactive mental health interventions tailored to individual patient trajectories.

Moreover, Kannan, Smith, and Chen [25] highlight the development of predictive modeling frameworks specifically designed to forecast illness progression in bipolar disorder. By continuously analyzing user-entered daily data, AI systems can not only monitor current mood states but also generate risk profiles, alerting users and healthcare providers to potential future episodes. This approach contributes to real-time mental health management while also addressing critical ethical concerns such as data privacy, patient consent, and methodological transparency challenges that have historically limited the widespread adoption of AI in mental health research.

The focus on structured, self-reported data thus offers a promising pathway for developing predictive digital health applications that are both user-friendly and clinically relevant, supporting patients' autonomy without requiring constant external monitoring or intrusive data collection.

2.4.2 Scientific Evidence for AI-Based Mood Prediction

Recent studies have shown that artificial intelligence (AI) techniques, particularly deep learning models, can predict mood fluctuations in people with bipolar disorder (BD) with over 80% precision using structured historical data such as mood logs, behavioral patterns, and daily routines [8]. AI-powered sleep monitoring further enhances these predictive capabilities, as irregular sleep patterns are strongly correlated with impending manic or depressive episodes [12].

Among the most commonly applied machine learning approaches are supervised learning algorithms, which are trained on labeled datasets to recognize patterns and classify mental states. For instance, Support Vector Machines (SVMs) are effective in separating complex mood categories by identifying optimal decision boundaries, making them suitable for classifying emotional states based on self-reported metrics. Similarly, Random Forests, which aggregate the results of multiple decision trees, perform well in noisy or incomplete data sets, conditions frequently encountered in digital mental health applications [26].

In addition to static classification models, Recurrent Neural Networks (RNNs) offer distinct advantages in analyzing time-series data. These models retain memory of previous inputs, allowing them to detect sequential patterns in mood fluctuations over time. This temporal modeling is particularly valuable for forecasting episodes before they escalate.

To improve overall predictive stability, ensemble learning methods combine the strengths of multiple algorithms, enhancing reliability and robustness especially in managing the subjectivity inherent in self-reported mental health data [26].

More recently, Transformer-based architectures have emerged as powerful alternatives to traditional sequential models. Unlike RNNs, which process data step-by-step, Transformers evaluate entire sequences simultaneously using attention mechanisms that capture both local and global context. In mood prediction tasks, this has led to improved precision. For example, Feng, Liu, and Zhao [27] achieved an Area Under the Curve (AUC) of 0.83 in classifying affective states using a Transformer model trained on clinician-rated assessments and self-reported

inputs.

Collectively, these AI methodologies provide a strong scientific and technical foundation for the development of advanced mood-tracking systems. Their integration into mobile health applications supports not only the accurate forecasting of mood episodes but also the delivery of proactive, personalized mental health interventions. These capabilities align directly with the objectives of this thesis, which seeks to design an AI-driven mobile platform for real-time mood prediction and early intervention in individuals with bipolar disorder.

2.4.3 Implementation of AI-Based Mood Prediction in Mobile Applications

AI-powered mood-tracking applications are designed to support bipolar disorder(BD)management by providing real-time insights, triggering proactive interventions, and issuing automated alerts. Future work could focus on integrating multimodal AI models combining behavioral, biometric and cognitive data to improve predictive robustness and personalization.[9].

Despite the significant promise shown by AI-driven mood forecasting models, widespread clinical implementation remains limited. Key barriers include concerns surrounding data privacy, algorithmic bias, and the lack of standardized protocols for integrating AI insights into clinical workflows. Ensuring patient data security and complying with international privacy regulations, such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA), remain critical challenges.

Moreover, AI models often perform well in controlled research environments but encounter difficulties when exposed to the variability and complexity of real-world patient populations. Ensuring model transparency, fostering user trust, and developing explainable AI systems are essential to promote wider acceptance among patients, clinicians, and regulators.

The integration of AI-based mood tracking with medication adherence monitoring and caregiver support tools could enhance BD management outcomes sig-

nificantly. However, addressing these ethical, technical, and systemic challenges is essential to achieve successful real-world adoption of predictive mood tracking applications [28].

2.4.4 Limitations and Challenges of AI in Mood Prediction

While AI-based models show high potential for mood prediction, several studies and real-world implementations reveal notable limitations. For instance, some mood prediction apps that initially demonstrated strong performance in lab conditions failed to maintain accuracy in real-world use due to inconsistent user input and diverse behavioral patterns. A study by Barnett et al. [29] found that machine learning models trained on controlled datasets lost over 20% of their accuracy when applied to real-world patient data with incomplete mood logs and irregular usage patterns.

Additionally, AI models are vulnerable to overfitting, especially when trained on small or homogeneous samples. This limits their ability to generalize across diverse populations. Moreover, some early-stage apps such as ***Monarca*** a Danish app for patients with BD using smartphone sensors and manual mood input, faced criticism for poor user adherence and complex interfaces, leading to high dropout rates during clinical trials [30].

These examples highlight the importance of designing systems that are both technically robust and user-friendly, while accounting for real-world variability in behavior, device use, and data quality.

2.5 Scope and Limitations

This thesis delivers a thorough review of AI-based mood prediction methodologies and outlines key technical, ethical, and regulatory considerations relevant to their design and application. However, several activities and deliverables are intentionally excluded from the scope of this research:

- **Clinical validation (primary research):** No new human-subject research, patient trials, or real-world clinical studies are conducted by the author to evaluate model efficacy, safety, or diagnostic accuracy. Existing clinical findings are discussed in the literature review, but no additional primary data collection or experimental trials are performed.
- **Production-level system development:** The implementation of a fully operational software application, including back-end integration, front-end optimization, security hardening, and deployment infrastructure, is not within the scope of this thesis.
- **Longitudinal and economic evaluation:** Extended follow-up studies measuring long-term patient outcomes, cost-effectiveness analyses, and health economics modeling are not addressed.
- **End-user training materials and deployment logistics:**
  - Designing instructional content and conducting usability testing
  - Planning deployment and establishing support workflows

Chapter 3

Methodology

3.1 Research and Design Methodology

The methodology for designing the Predictive Mood Tracking App for Bipolar Disorder Patients follows a user-centered design (UCD) approach, ensuring that the application caters to the unique needs of individuals managing bipolar disorder. The design methodology is built on a combination of clinical research, psychological principles, and user feedback to create an app that is not only clinically effective but also emotionally supportive and accessible. This study presents a conceptual design for a bipolar mood-tracking application informed by literature-based analysis and benchmarking of existing features. As no clinical testing or real-user involvement was conducted at this stage, the proposed design remains theoretical and will require future validation through direct user feedback and clinical collaboration.

3.1.1 Key Features of the Design Methodology for the App

User-Centered Design (UCD) The app was designed with the primary goal of addressing the specific needs of individuals with bipolar disorder. A user-centered approach was employed, focusing on understanding the challenges users face, including cognitive overload, emotional instability, and the need for proac-

tive mood management. The app was designed to be intuitive and easy to navigate, particularly for individuals who may experience fluctuating cognitive abilities during manic or depressive episodes. The design takes into account the need for simple, straightforward interactions and minimal cognitive load, essential for individuals with mental health challenges [31].

Emotional and Cognitive Accessibility A key challenge in designing an app for people with bipolar disorder is balancing emotional engagement with functional simplicity. The design prioritizes emotional support by incorporating affirming language, motivational messages, and a calming color palette (teal and blue) that help reduce anxiety and support emotional regulation. Simultaneously, the design remains user-friendly, ensuring that the app's core features—such as mood tracking, medication adherence, and self-care reminders—are easily accessible. Research has shown that emotional design is essential for apps aimed at improving mental health, as it enhances engagement and user well-being [32].

Predictive Mood Monitoring and AI Integration A core feature of the app is its AI-driven predictive mood monitoring, which forecasts potential mood fluctuations based on historical data. The Mood Log, Calendar, and To-Do Suggestions sections incorporate AI to anticipate changes in emotional states and provide timely reminders or suggestions. Research by Kannan, Smith, and Chen [25] emphasizes the role of AI models in improving the accuracy of mood predictions, making it easier for individuals to manage mood episodes proactively.

Incorporating Motivational Messaging and Activity Suggestions Alongside mood prediction, the app generates AI-driven motivational messages and personalized activity suggestions. Based on the user's mood, the app might provide supportive messages like "You're doing great, just check in with yourself," or suggest tasks suited to the user's emotional state. Signorelli et al. [33] highlight how personalized messaging enhances user engagement and emotional resilience, which are essential for improving adherence to mental health management routines.

Privacy and Ethical Considerations Given the sensitive nature of the data involved (e.g., mood logs, medication adherence), the app's design prioritizes data privacy and security. The app includes clear privacy policies and consent forms to ensure users understand how their data will be used. Users have complete control over what data they share with healthcare providers or trusted support persons, ensuring transparency and compliance with privacy regulations (e.g., GDPR, HIPAA).

Sustainability and Long-Term Engagement Encouraging long-term use is a significant challenge when designing mental health apps. To address this, the app incorporates gamification features, such as a points-based reward system for completing daily mood assessments and staying consistent with medication tracking. These features help build healthy habits and encourage ongoing engagement. Additionally, the app is designed to provide real-time feedback and positive reinforcement, which are crucial for maintaining motivation and adherence to mental health management routines over time [34].

3.2 Understanding User Needs

The foundation of the app's design lies in a deep understanding of the needs, challenges, and preferences of individuals living with bipolar disorder. This phase involved reviewing existing literature on bipolar disorder, current treatment approaches, and digital health solutions to identify gaps in existing services and ensure the app would provide meaningful support for users.

3.2.1 Literature Review and Clinical Insights

To understand the complexities of bipolar disorder and its impact on daily life, an extensive literature review was conducted, focusing on clinical research, treatment methods, and the challenges that patients face in managing their condition. Key findings revealed that mood tracking, medication adherence, and emotional regulation are critical factors in managing bipolar disorder [35].

Studies emphasize that mood tracking is particularly important for early detection of mood shifts, which can help mitigate episodes of mania or depression [36]. Moreover, individuals with bipolar disorder often experience difficulties with consistency in self-monitoring and adherence to treatment plans, which can exacerbate mood fluctuations [37].

Based on these findings, the app's design was centered around user empowerment, encouraging users to track their moods daily and providing tools to help them manage their medication adherence.

3.2.2 Examining Existing Solutions

An analysis of existing mood-tracking and mental health apps also informed the design. While many apps provided basic mood tracking and medication reminders, there was a noticeable lack of personalized support and tools for predictive mood monitoring. Most apps fail to proactively help users anticipate mood shifts or provide real-time emotional support. These gaps were identified as opportunities to enhance the user experience and provide a more holistic approach to managing bipolar disorder.

Furthermore, many existing apps did not incorporate sufficient privacy controls or failed to address the emotional sensitivities of users who may feel anxious about tracking their moods or sharing sensitive data. As a result, user-centered design principles were emphasized to ensure that the app offers a safe and supportive environment.

3.2.3 Identifying Key User Needs

From the literature review and analysis of existing solutions, the following key needs of users with bipolar disorder were identified:

- **Proactive Mood Monitoring:** Users need tools that predict mood fluctuations and give them enough time to take preventive actions.
- **Emotional Support:** Offering motivational messaging and personalized

activity suggestions can provide essential emotional guidance during mood shifts.

- **Medication Adherence:** Ensuring that users take their prescribed medications on time is crucial for managing the disorder effectively.
- **User Control and Privacy:** Users should have complete control over their data and the ability to share it only with trusted individuals, such as healthcare providers and family members.
- **Simple, Intuitive Design:** Given the cognitive challenges that can accompany mood episodes, the app needs to be easy to navigate and not overwhelm users with complex features.

3.3 Literature-Based Requirement Gathering

The requirement gathering process for the Predictive Mood Tracking App was conducted through a literature-based approach, rather than through direct user or clinician input. This conceptual design phase focused on ensuring that the proposed features were both clinically relevant and emotionally supportive for individuals with bipolar disorder. Key functional and emotional requirements were identified through an extensive review of peer-reviewed literature, clinical guidelines, existing mobile health applications, and expert recommendations in the field of digital mental health. Insights derived from academic sources, mental health journals, and published evaluations of similar applications informed the selection of core functionalities. These include daily mood self-assessments, AI-driven mood predictions, and personalized suggestions for emotional self-regulation. While the current design is based on secondary data

3.3.1 Understanding Bipolar Disorder: Key Challenges and Requirements

A comprehensive literature review was conducted to identify the unique challenges and needs of individuals with bipolar disorder, with a specific focus on how technology can assist in managing the condition. The review concentrated on the following key areas:

Mood Tracking for Early Intervention and Self-Management Research highlights the importance of mood tracking as a critical tool for identifying early signs of mood shifts, facilitating early intervention, and supporting better self-management for individuals with bipolar disorder. Studies have demonstrated that consistent mood tracking enables users to detect and anticipate both manic and depressive episodes, enhancing their ability to manage their emotional state effectively [35].

Medication Adherence One of the significant challenges faced by individuals with bipolar disorder is medication adherence. Non-compliance with treatment regimens is common and can lead to mood instability and relapse. Literature underscores the need for tracking medications and setting reminders to ensure consistency with prescribed treatments. Effective adherence plays a vital role in stabilizing mood and preventing episodes [37].

Emotional Regulation and Personalized Feedback Managing emotional fluctuations is a central challenge for individuals with bipolar disorder. Personalized feedback, such as motivational messaging and activity suggestions, plays a crucial role in helping users regulate their emotions. Research indicates that providing context-sensitive support based on emotional state can foster emotional resilience and help manage mood swings, contributing to improved long-term mental health outcomes [26].

3.3.2 Existing Apps Analysis

An analysis of existing mood-tracking apps and other mental health applications offered valuable insights into the strengths and weaknesses of current solutions. The review highlighted several critical gaps in the market, such as:

- The lack of predictive mood monitoring, which allows users to anticipate and prepare for emotional shifts.
- Insufficient personalized emotional support or real-time feedback tailored to users' current emotional states.
- Limited collaboration features for patients, doctors, and trusted support persons to monitor mood trends and medication adherence together.

By identifying these gaps, the app's design aimed to address unmet needs by incorporating AI-based predictive tools, personalized motivational messaging, and ensuring features that support collaborative care between patients, healthcare providers, and trusted support persons.

3.3.3 Defining User Requirements and Predictive System Design

The design of the *Predictive Mood Tracking App* was guided by a literature-based requirements gathering process, informed by clinical studies, digital mental health research, and critical analysis of existing bipolar disorder applications. This conceptual approach was necessary due to the absence of real-world user testing or direct involvement of clinicians at this stage of development. Nevertheless, the resulting framework reflects current best practices in user-centered design and mental health technology development.

3.3.4 User Needs and Functional Priorities

Through the synthesis of scientific literature, clinical guidelines, and app benchmarking studies, several core user needs were identified as essential for the app's success in supporting individuals with bipolar disorder [18, 25, 38]:

- **Mood tracking:** An intuitive, low-effort interface for logging daily mood fluctuations and associated symptoms.
- **Personalized support:** AI-powered motivational messages and activity suggestions tailored to the user's current or predicted emotional state.
- **Proactive intervention:** Predictive alerts that enable users to anticipate and respond to potential manic or depressive episodes in advance.
- **Medication adherence:** A reminder system that helps users track and follow prescribed treatments consistently.
- **Collaborative care:** Options for sharing selected data with healthcare providers or trusted support persons to promote coordinated care.
- **Privacy and autonomy:** Transparent data-sharing controls and user governance over all personal information.

These features were prioritized to ensure emotional accessibility, clinical relevance, and long-term engagement. They also reflect common shortcomings observed in current bipolar disorder tracking tools, which often lack personalization, predictive capability, and integration with broader care networks [19, 23].

3.3.5 Conceptual Design of the Predictive System

In alignment with these user needs, the app includes a conceptual AI-driven mood prediction system designed to identify patterns in mood, behavior, and routine. At an initial level, this system could use a rule-based decision tree to generate alerts. For example, if a user records three consecutive days of low mood, disrupted sleep (<4 hours), and missed medication, the system would classify the user as "at risk of a depressive episode" and offer supportive feedback or intervention suggestions. Conversely, high activity levels, elevated mood entries, and reduced sleep may indicate the onset of mania.

This structured, explainable model offers a transparent and ethical foundation for mood prediction—especially important in sensitive mental health contexts [26]. As the system matures, more advanced approaches such as Recurrent

Neural Networks (RNNs) or Transformer-based models could be incorporated to detect complex temporal and behavioral patterns. These models, trained on anonymized longitudinal data, would enhance the system's predictive accuracy and personalization [27, 39].

By embedding the predictive logic directly into core app features—such as the *Mood Log*, *Calendar*, and *To-Do Suggestions*—the system helps users manage emotional fluctuations proactively and maintain greater mood stability.

3.4 Implementation of the Predictive System Within the App

Building on the identified user needs and conceptual model, the following sections explain how the predictive mood system is integrated into the app's features, how the AI component functions, and what ethical considerations were addressed.

3.4.1 Integration with App Features

The predictive mood monitoring system is integrated with several key features of the app, including the Mood Log, Calendar, and To-Do Suggestions. This integration ensures that users receive personalized and timely support across different areas of the app.

- **Mood Log:** The Mood Log feature prompts users to log their mood daily, providing the app with real-time data that can be used for predictions. Based on the historical data collected, users are shown a personalized reminder each day regarding their predicted emotional state (e.g., mania or depression). This functionality aligns with research showing that consistent mood tracking is critical for managing bipolar disorder [35].

- **Calendar:** The Calendar feature provides users with a visual representation of predicted mood cycles using color-coding. Manic days are represented in orange, while depressive days are marked in blue. This allows

users to visualize their emotional trajectory, which is aligned with findings that suggest early detection and mood forecasting improve mood management and reduce the severity of episodes [26].

- **To-Do Suggestions:** Based on the predicted mood state, the To-Do Suggestions feature offers personalized activity recommendations. These recommendations are designed to help users regulate their emotional state by providing suitable activities. For example, users experiencing low mood might be encouraged to engage in calming activities, such as deep breathing exercises or taking a walk. Conversely, users experiencing high energy (during a manic episode) might be guided to engage in focused activities like journaling or creative tasks. This dynamic approach to emotional regulation is supported by research on AI-driven personalized feedback [33].

3.4.2 AI Model and Data Processing

While the technical implementation of the AI model falls outside the scope of this design phase, a conceptual framework for its functionality is proposed to demonstrate how mood prediction could be achieved. The system is envisioned to rely on machine learning algorithms that analyze users' mood trends, sleep quality, energy levels, and medication adherence, as suggested by recent studies on multimodal mood analysis [9, 25].

At an initial stage, the model could be implemented using a rule-based decision tree that categorizes user states based on predefined thresholds. For example, if a user logs "low mood" for three consecutive days, reports less than four hours of sleep, and indicates missed medication, the system could classify the user as "at risk of a depressive episode" and trigger a supportive alert. Conversely, consistent high energy, reduced need for sleep, and increased activity logging might indicate a potential manic episode. This type of interpretable logic aligns with recommendations for ethical and explainable AI in mental health contexts [28].

As the system evolves, more advanced models such as Recurrent Neural Net-

works (RNNs) or Transformer-based architectures could be integrated to recognize more complex temporal patterns and subtle changes in behavior over time. These models have been shown to improve mood state prediction in bipolar disorder when trained on longitudinal, user-generated data [27, 26].

The AI engine would then deliver early warning signals via in-app prompts, motivational messages, and adaptive suggestions for mood regulation. By integrating this predictive feedback into the *Mood Log*, *Calendar*, and *To-Do Suggestions* features, the app supports users in taking proactive steps to manage emotional fluctuations and maintain long-term stability, an approach emphasized in current digital psychiatry frameworks [20].

3.4.3 Ethical Considerations and User Trust

Given the sensitive nature of emotional and behavioral data, ethical considerations were central to the design of the predictive mood monitoring system. The app is structured to uphold user autonomy by offering full control over personal information, including options for users to manage what data they share, with whom, and under what conditions. Sharing features are entirely optional and are governed by user-initiated consent settings that comply with data protection standards such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA) [28].

To enhance transparency, users are presented with clear information regarding how their data will be used particularly in the predictive system. Informed consent is built into the onboarding process and can be revised at any time through in-app settings. The app ensures that users remain in charge of their digital health footprint, with options to update, revoke, or delete their data completely.

A major ethical concern in AI-based mental health tools is the risk of user anxiety triggered by predictive alerts. To address this, the system is designed to communicate predictions in a gentle, supportive manner. Rather than framing mood forecasts as clinical warnings, the app provides empowering suggestions and motivational prompts aligned with the user's emotional state. For example, instead of stating *"You may be entering a depressive phase"*, the app might

suggest, *"This might be a good day to focus on rest and self-care"* [40].

This approach aims to foster trust, minimize emotional distress, and reinforce the user's sense of control and agency in managing their mental health.

3.5 Prototyping and UI/UX Development

The process of prototyping and UI/UX development was essential in turning the initial design concepts into a tangible, interactive app interface. This phase allowed for the visualization and refinement of the app's features and user flows, ensuring that the design would be both functional and emotionally supportive for individuals with bipolar disorder. Prototyping played a critical role in iterating on the design, incorporating user feedback, and ensuring the app's usability.

3.5.1 Prototyping Tools and Design Iterations

To create the app's user interface (UI) and user experience (UX), Figma was used as the primary tool for wireframing and prototyping. Figma allowed for rapid iterations of the design, enabling real-time collaboration and feedback from stakeholders. The prototyping process involved several key steps:

- **Wireframing:** The initial wireframes were developed to outline the basic structure of the app, focusing on layout, content arrangement, and navigation flow. Wireframes served as the first visual representation of the app, helping to define the app's core features, including the Mood Log, Calendar, Medication Tracker, and My Team (trusted support persons).

- **User Flows:** The user flows were mapped out to ensure a seamless experience for users as they navigated between the different sections of the app. The user flows emphasized the importance of simplicity and accessibility, particularly for users who might experience cognitive challenges during manic or depressive episodes.

- **Interactive Prototypes:** Interactive prototypes were created to simulate the actual app experience. These prototypes included clickable buttons,

transitions between screens, and dynamic features such as mood tracking and medication reminders. This allowed for testing the user journey and refining the interface for ease of use, user engagement, and emotional support.

3.5.2 Design Principles for UI/UX

The design of the app was guided by key UI/UX principles to ensure the app would be user-friendly, accessible, and emotionally supportive. These principles were informed by both clinical research on bipolar disorder and best practices in digital mental health design:

- **Simplicity and Clarity:** The UI was designed to be as simple and intuitive as possible, reducing cognitive load and avoiding overwhelming the user. Large, readable fonts, high-contrast color schemes, and clear icons were used to enhance legibility and ease of navigation. Minimalist design principles were adopted to ensure that users could easily access the app's core features, even during mood fluctuations [31].

- **Emotional Support:** A primary consideration in the UI design was ensuring that the app provided emotional support. The color palette, consisting of soft teal green and blue, was chosen for its calming effects, reducing anxiety and promoting emotional regulation [41]. Additionally, affirmative language was incorporated throughout the app, with messages such as "You're doing great, just check in with yourself," designed to provide users with emotional reassurance.

- **Consistency and Familiarity:** The design utilized familiar UI patterns, ensuring that the layout and navigation were consistent with what users typically encounter in digital applications. The tab bar navigation at the bottom of the screen provides quick access to essential features, and iconography was kept simple to help users easily recognize key functions [18].

- **Accessibility:** Special attention was given to accessibility features, ensuring that the app was usable by individuals with various disabilities. This includes screen reader compatibility, adjustable text sizes, and high-contrast visuals. These features are critical for individuals with bipolar disorder, who may experience visual or cognitive difficulties during mood episodes.

3.5.3 Visual Design and User Interaction

The visual design was created with the goal of fostering a positive user experience. The color scheme was selected to evoke feelings of calmness and emotional balance, while typography choices ensured readability and reduced stress for users. Iconography was designed to be simple and intuitive, allowing users to easily understand app features and navigate between sections. Motion design was used sparingly, offering gentle transitions to guide users through the app without causing distraction.

Incorporating user feedback from informal testing and professional input, the app's interface was optimized to ensure it met the needs of individuals with bipolar disorder, enabling them to track their mood, manage their medication, and receive support in a clear and emotionally supportive environment.

3.6 Ethical and Privacy Considerations

Given the sensitive nature of the data involved, ethical considerations and privacy were central to the design of the Predictive Mood Tracking App. The app collects and processes personal health data, including mood entries, medication adherence, and emotional feedback, which must be handled with the utmost care and in compliance with relevant regulations. In this section, we discuss the ethical principles guiding the design and the privacy measures implemented to ensure that user data is securely stored, shared, and protected.

3.6.1 User Consent and Transparency

A foundational principle in the app's design is that users have full control over their data. Clear informed consent processes were integrated to ensure that users understand how their data will be used. Upon registering, users are presented with a privacy policy and terms of service that explain the types of data collected, how it will be used (e.g., mood tracking, medication adherence), and who can access it (e.g., healthcare providers, trusted support persons). This ensures that users are fully aware of their rights and the scope of data collection before they begin using the app.

Informed consent also extends to the predictive mood tracking features, where users are made aware that their emotional data is being used to predict mood states and provide personalized recommendations. They are given the option to opt-in or out of certain features, maintaining user autonomy while ensuring that the app serves its intended purpose of emotional support.

3.6.2 Data Security and Encryption

To protect user privacy and comply with data protection regulations (e.g., GDPR and HIPAA), all personal and health data is encrypted both during transmission and at rest. The app uses secure server infrastructure to store user data, ensuring that unauthorized parties cannot access sensitive information. The data storage system is designed to ensure that only authorized users (e.g., the patient, healthcare providers, or trusted support persons with user consent) have access to specific data sets.

User authentication features, such as two-factor authentication (2FA) and password protection, have been incorporated to safeguard accounts against unauthorized access. This ensures that users' data is only accessible to them and individuals they trust, thereby enhancing the app's security and users' trust in the system.

3.6.3 Data Sharing and User Control

The app enables users to securely share their mood and medication data with healthcare providers or trusted support persons (e.g., family members, partners) to support collaborative care. This sharing is entirely voluntary and is managed through explicit user-controlled consent settings. Users can choose what specific data they wish to share, who is permitted to access it, and the duration of access.

For example, if a user chooses to share their data with a psychiatrist or family member, they can do so by selecting the individual from their approved contact list within the app's settings. The selected person would then be granted limited access to view mood trends and medication adherence reports. All shared data is encrypted during transmission and protected through secure sharing protocols, ensuring compliance with privacy regulations such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA) [28].

This functionality supports the collaborative care model, which has been shown to improve treatment outcomes and patient adherence in individuals managing bipolar disorder.

The app's privacy settings also empower users to modify or revoke sharing permissions at any time, reinforcing user autonomy and control. Additionally, users may permanently delete their data from the system, honoring the "right to be forgotten" under current privacy legislation.

3.6.4 Ethical Use of Predictive Data

The use of AI-based predictive mood monitoring raises important ethical questions, particularly around the potential for misuse or misinterpretation of predictive data. To address these concerns, the app is designed to present predictions and suggestions in a non-alarming manner, ensuring that users understand these are projections and not absolute predictions. Users are encouraged to consult with healthcare providers if they have concerns or if the predictive alerts seem out of alignment with their experiences.

Additionally, the app includes supportive language and gentle reminders to encourage users to engage with predictions constructively rather than anxiously. The aim is to empower users, not to increase their emotional distress, and to foster trust in the system.

3.6.5 Compliance with Regulations

The app adheres to the highest standards of data privacy and security to comply with global privacy regulations such as the General Data Protection Regulation (GDPR) in Europe and the Health Insurance Portability and Accountability Act (HIPAA) in the United States. These regulations require that users' personal and health data be handled with the utmost care and be protected against unauthorized access or misuse.

To ensure compliance, the app includes mechanisms for data anonymization, data encryption, and clear user consent processes. All data handling practices are regularly reviewed to ensure they meet the standards set by these regulations, reinforcing the app's commitment to maintaining user privacy and confidentiality.

Chapter 4

Design and Concept

4.1 App Design Overview and Philosophy

This chapter presents a high-fidelity prototype of the Predictive Mood Tracking App. While the visual design and functional architecture are conceptually detailed, no actual software development, database integration, or clinical deployment has been implemented. The system remains at the prototype stage and serves as a simulation of the intended features and user interactions. The design philosophy underpinning this application is grounded in user-centered, inclusive, and clinically informed principles, with the overarching goal of enhancing the daily lived experiences of individuals with bipolar disorder. From its inception, the app was conceived as a digital support system not only for patients but also for their healthcare providers and trusted support persons (e.g., family members, partners, or close friends). This triadic care model was designed to bridge existing communication gaps and facilitate timely emotional and medical support through shared access to mood trends, medication adherence records, and emotional state insights.

4.1.1 Compliance with Human Interface Guidelines

The application's design is informed by Apple's Human Interface Guidelines (HIG), which emphasize clarity, depth, and deference to user content. These

principles have been consistently applied throughout the interface to create a system that feels intuitive, familiar, and emotionally supportive for individuals managing bipolar disorder. In accordance with HIG's clarity principle, all visual elements—including typography, color use, iconography, and layout—are designed to communicate information efficiently and with minimal cognitive demand. Depth is achieved through subtle layering, shadows, and transitions, giving users a clear sense of spatial hierarchy without causing distraction. Deference is maintained by ensuring the interface never overwhelms the content, keeping users focused on their mood entries, medication adherence, and supportive messages.

In addition, the use of spacing, alignment, and responsiveness across screen sizes follows Apple's recommended design grids, ensuring visual consistency and accessibility. In alignment with HIG, the application not only provides a polished and professional visual language, but also adheres to recognized industry standards, facilitating user trust and engagement, particularly for users already familiar with iOS environments. This grounding in Apple's HIG ensures that the app supports smooth navigation, low-friction interaction, and a calm emotional experience, all of which are essential in mental health applications.

4.1.2 Multi-User Role Integration

Upon first launch, users are asked to select whether they are a patient or a doctor. Each role is directed to a customized interface: Patients can track symptoms, mood cycles, and medication adherence, while doctors receive tools to remotely monitor these patterns, schedule appointments, and communicate via in-app messaging. Additionally, patients can invite a trusted support person who receives limited, yet meaningful, access to mood and adherence data. This collaborative setup aligns with shared care models in mental health [42], which emphasize improved outcomes when patients, caregivers, and clinicians participate in joint monitoring and decision-making.

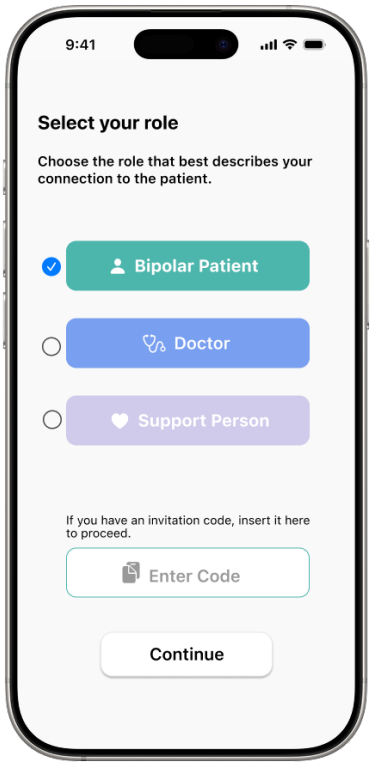


Figure 4.1: Multi-User Role Integration

4.1.3 Color Palette for Psychological Safety

The app uses a calming and psychologically safe color palette teal green (#4DB6AC) and royal blue (#3478F5) based on established principles in color psychology. Research shows that blue and green tones promote emotional regulation, reduce anxiety, and increase perceived trust [41, 43]. As Küller et al. [41] and Kaya and Epps [43] emphasize, "Color palettes that prioritize emotional balance and low arousal are best suited for individuals with affective disorders, particularly when those palettes incorporate blue, green, and neutral tones." Blue is often associated with calmness and cognitive clarity, while green evokes emotional stability and reassurance.

4.1.4 Minimalist and Affirmative Interface

The app's visual design follows minimalist principles with a clean background to reduce visual overload and support readability. Large, accessible fonts, and intuitive icons help guide the user. Affirmative language is used throughout the

app, for example, prompts such as “You’re doing great” or “No rush, just check in with yourself”, to create an emotionally safe and encouraging environment.

Accessibility and Inclusivity Considerations Accessibility and inclusivity were foundational considerations in the design process, ensuring that users of varying cognitive, emotional, and digital literacy levels could engage with the app effectively. The interface was intentionally designed with large touch targets, high contrast ratios, and readable typography, in alignment with accessibility guidelines for users with visual strain or motor difficulties. Navigation is streamlined and linear to accommodate users who may experience cognitive fatigue, especially during depressive or manic episodes.

Additionally, content is structured to minimize overwhelming the user. By using soft color cues and simple progress indicators, the interface helps reduce information overload while still providing clarity. Language used throughout the app is friendly, non-judgmental, and emotionally affirming, further enhancing accessibility for individuals who may be in emotionally vulnerable states. These efforts aim to create a safe, inclusive, and emotionally stable experience for all users, regardless of their level of technological proficiency or emotional state at the time of use.

4.1.5 User-Centered Design Approach

The design of the application adheres to a user-centered design (UCD) approach, which prioritizes understanding the needs, behaviors, and emotional states of the target users. This approach is particularly important when designing for individuals with bipolar disorder, as their symptoms fluctuate and require a highly adaptable and responsive interface. Although real-world user tests were not conducted during this phase, the design process was informed by established clinical frameworks, psychological theories, and empirical research on bipolar disorder, ensuring that the application is clinically relevant and emotionally supportive.

Informed by Clinical Tools The daily mood assessment questions within the app were adapted from validated clinical tools such as the Internal State Scale (ISS), the Mood Disorder Questionnaire (MDQ) and the diagnostic criteria of the DSM-5 [2]. These tools are widely used in clinical settings to assess mood and symptom severity. The seven self-report questions on the app were designed to capture critical mood symptoms, such as depression, mania, sleep disturbance, and cognitive function, while being simplified for daily use on a mobile platform. The integration of these clinical tools ensures that the app remains grounded in established diagnostic practices and is suitable for clinical monitoring.

Promoting Engagement through Gamification and Routine Building To foster consistent participation and enhance user motivation, the app incorporates gamification elements grounded in behavioral design principles. These include a point-based reward system, progress tracking through visual streaks or calendars, and the unlocking of motivational badges after achieving milestones such as consecutive mood logs or medication adherence.

Such features are especially beneficial for individuals with bipolar disorder, who may face fluctuating levels of energy, motivation, and routine stability. By offering tangible incentives and visual feedback, the app supports habit formation and encourages daily self monitoring even during periods of low engagement. For example, completing a week of consistent mood logging may prompt a personalized badge and a motivational message tailored to the user's emotional state.

Research highlights that gamified strategies can improve adherence and engagement in digital health contexts, particularly when combined with meaningful feedback and user-centered rewards [34, 44]. By integrating these mechanisms, the app not only sustains user interest but also reinforces emotional self-awareness and proactive health behaviors.

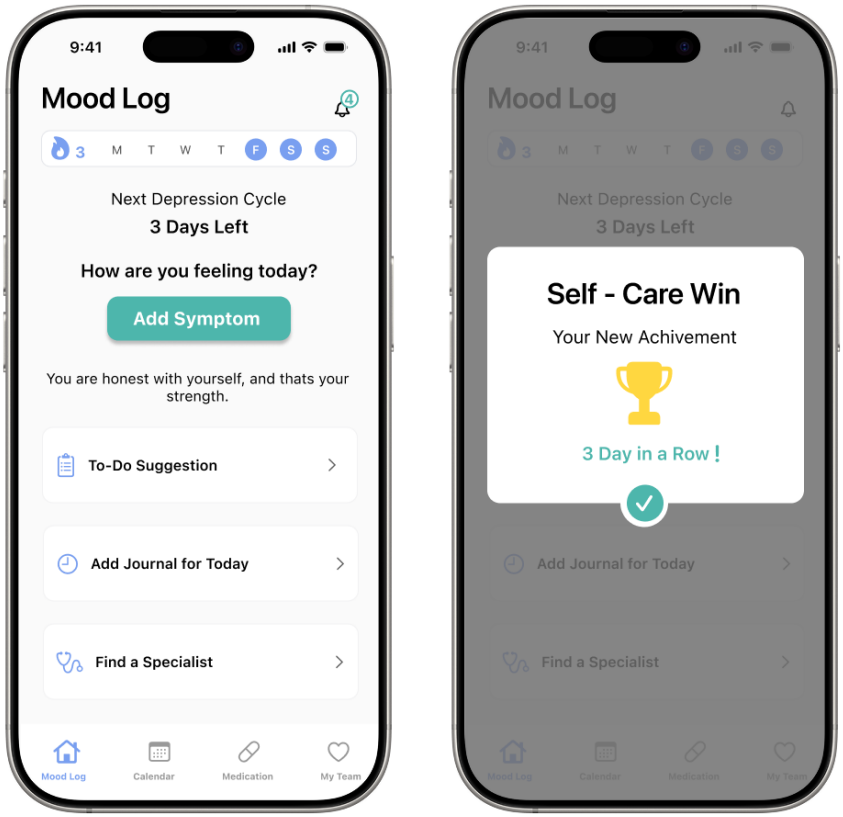


Figure 4.2: Promoting Engagement through Gamification

Emotional Support through Visual and Linguistic Design The design emphasizes emotional support through visual and linguistic elements that promote psychological safety. The color palette, calming shades of blue and green, was chosen based on research in color psychology, suggesting that these hues reduce anxiety and promote emotional regulation in affective disorders [41, 43]. Affirmative language and gentle prompts (e.g., "You're doing great, just check in with yourself") create a non-judgmental, supportive environment that encourages consistent engagement.

Multi-User Collaboration and Shared Care The app supports a triadic care model, integrating the patient, their healthcare provider, and trusted support persons (e.g., family members or partners). This setup is inspired by shared-

care research in mental health, which shows improved outcomes when patients, caregivers, and clinicians share monitoring and decision-making responsibilities [42]. Through real-time monitoring of mood trends, medication adherence, and emotional states, the app facilitates improved communication and collaboration, fostering a holistic approach to bipolar disorder management.

4.1.6 Emotional and Behavioral Design Principles

The emotional and behavioral design principles of the application are grounded in psychological theories and clinical frameworks aimed at supporting users in managing their mood disorders. Recognizing the emotional complexity of individuals with bipolar disorder, the app's design emphasizes emotional regulation, user engagement, and consistent behavior tracking.

One of the key aspects of the design is the incorporation of motivational language and visual cues that provide emotional support throughout the user's journey. Motivational sentences are strategically placed above each daily symptom check-in question to encourage users to engage with the app without feeling judged or overwhelmed. These affirmations, such as "You're doing great, just check in with yourself", are intended to foster a sense of psychological safety, reduce emotional stress, and enhance user engagement. This approach aligns with research showing that positive reinforcement and gentle prompts are effective in maintaining motivation for individuals managing chronic conditions [34].

Additionally, the app offers personalized features, including To Do suggestion, mood tracking, and AI-generated to-do suggestions. These suggestions are tailored to the user's current mood, guiding them to perform gentle activities such as drinking water, taking a short walk, or writing a journal entry. This approach not only supports emotional regulation but also encourages users to develop a routine of self-care and mindfulness. The inclusion of these features, in combination with consistent mood tracking, fosters a sense of agency and control over emotional states, which is critical for individuals with bipolar disorder.

The app's design also includes a report generation feature that enables users to track their mood fluctuations, medication adherence, and symptom history

over time. By offering daily, monthly, and yearly reports, the app facilitates a comprehensive understanding of mood patterns, which is crucial for both the user and their healthcare providers. This feature enhances collaborative care and supports clinical decision-making by providing a clear, data-driven view of the user's emotional and behavioral progress.

Incorporating these emotional and behavioral design principles ensures that the app not only serves as a tool for tracking symptoms but also functions as a supportive, engaging, and motivating companion for individuals managing bipolar disorder.

4.1.7 Color System and Psychological Rationale

The color palette of the proposed application plays a fundamental role in shaping the emotional experience of users, particularly individuals with bipolar disorder, for whom mood stability and reduced cognitive load are critical. The visual design was informed by principles of color psychology, emotional perception, and accessibility, aiming to create an interface that is not only functional but also emotionally supportive and calming.

Primary Colors

- **Soft Green (#4DB6AC)**

  Used across primary action buttons and success indicators, this desaturated green is associated with emotional balance, psychological harmony, and natural healing. According to Kaya and Epps [43], green tones reduce stress and foster a sense of stability—particularly beneficial for individuals experiencing agitation or depressive fatigue. Its muted appearance contributes to a reassuring and grounding atmosphere during mood fluctuations.

- **Soft Blue (#799FF0)**

  Applied to links, navigation elements, and prompts for user engagement, this gentle blue promotes calmness, trust, and cognitive clarity. As supported by Wexner [45], blue hues are perceived as emotionally soothing

and capable of lowering heart rate, making them ideal for users managing anxiety or emotional turbulence.

- **Muted Pink (#D0CAEB)**
  Reserved for trusted support person interactions, this soft pink was chosen for its associations with empathy, emotional warmth, and care. It creates a gentle and compassionate tone that encourages interpersonal connection without overstimulation.

Background Color

- **Off-White (#FAFAFA)**
  Used globally as the application's background, this soft off-white improves readability and reduces eye strain. It avoids the harshness of pure white while maintaining a high contrast level that supports clarity and comfort.

Clinical Justification and Emotional Safety Color perception acts as a non-verbal cue in emotional processing. Individuals with bipolar disorder often exhibit heightened emotional sensitivity, particularly during depressive or manic episodes. Studies show that specific color environments can regulate behavioral responses and reduce emotional arousal [43, 46]. As noted by Kaya and Epps [43], "Color palettes that prioritize emotional balance and low arousal are best suited for individuals with affective disorders, particularly when those palettes incorporate blue, green, and neutral tones."

The selected palette aligns with these findings, emphasizing calming, pastel-based hues that support psychological containment, clarity, and stability. In mobile health (mHealth) applications, a visually soothing and emotionally safe interface has been shown to reduce abandonment rates and improve long-term engagement [47]. Therefore, the deliberate exclusion of intense reds or highly saturated tones ensures that the design avoids overstimulation while supporting emotional regulation and comfort.

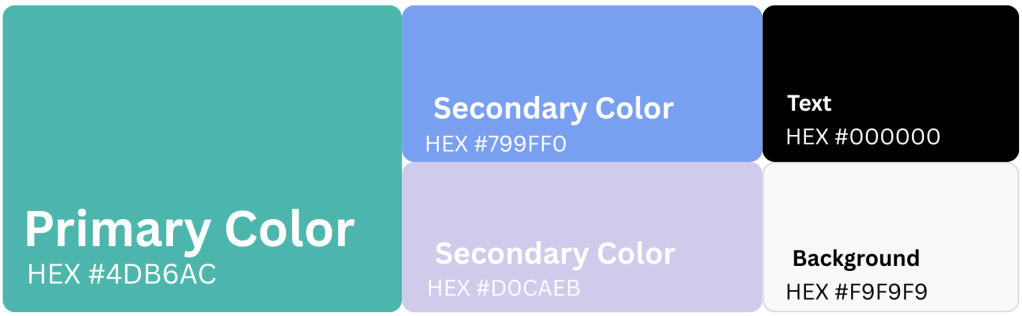


Figure 4.3: Color Palette

4.2 Wireframes and UI/UX Design

The wireframes and user interface (UI)/user experience (UX) design of the application were developed with strong emphasis on user-centered and evidence-based digital health principles, tailored specifically for individuals with bipolar disorder. Research in digital mental health highlights the importance of intuitive, emotionally supportive, and accessible design to foster engagement and improve clinical outcomes [18, 20, 44].



Figure 4.4: Wireframe Screens Representing Core User Flows

Wireframing Process and Iterative Design Wireframes were created in Figma to define the app's structure, layout, and functionality. This iterative design process allowed early identification of usability issues and incorporated multidisciplinary feedback from clinicians, designers, and developers [14]. Special attention was given to reducing cognitive load and simplifying user interaction, as individuals with bipolar disorder may experience emotional sensitivity and concentration difficulties [23, 22].

4.2.1 UI Design Elements

The user interface (UI) of the Predictive Mood Tracking App was designed with clarity, emotional safety, and accessibility in mind. Each visual and interactive element was carefully selected to support the cognitive and emotional needs of users with bipolar disorder. The following design elements reflect evidence-based digital health guidelines and user-centered design principles.

- **Typography:** Typography plays a central role in supporting usability and emotional comfort. The application uses the SF Pro typeface, a modern, highly legible system font developed for digital interfaces. Large font sizes (e.g., 32pt for main screen titles) ensure high readability and reduce visual strain an important consideration for users who may experience concentration difficulties during mood episodes. A consistent typographic hierarchy is maintained throughout, using bold and semibold weights for titles and buttons, and regular or light weights for descriptive text and captions. This structure enhances user orientation while maintaining a calm and unobtrusive visual tone. Affirmative messages and mood feedback are presented using gentle phrasing and soft weights to foster emotional safety without overwhelming users.

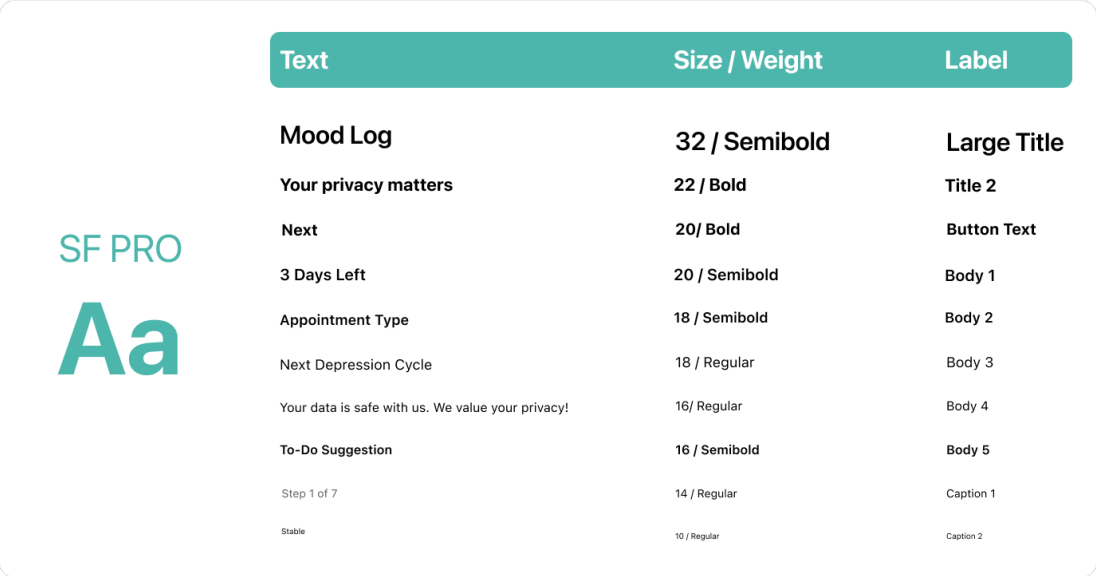


Figure 4.5: Typography hierarchy and font styles used across the app interface

- **Apple's HIG Compliance in Typography** In accordance with Apple's Human Interface Guidelines (HIG), the application adopts clean, legible, and scalable typefaces. Decorative or overly condensed fonts were deliberately avoided to ensure accessibility across various screen sizes and cognitive states. Font weights and spacing were chosen to maximize clarity, aligning with HIG's core principles of clarity, deference, and depth, ensuring that text remains functional without distracting from content.

- **Color Scheme:** The app's color palette is designed to be both psychologically supportive and functionally accessible, using muted tones to promote emotional regulation and reduce overstimulation in users with bipolar disorder.

  Primary Green (#4DB6AC): Used across key call-to-action elements for bipolar patients, this soft teal evokes emotional balance, calmness, and stability.

  Secondary Blue (#799FF0):Used for doctor dashboard buttons and indicators, blue conveys clarity and professionalism, while helping reduce anxiety.

Supportive Pink (#D0CAEB): Assigned to elements for trusted support persons, this soft lavender tone evokes empathy and reassurance without overstimulation.

Background Color (#F9F9F9): A light, neutral tone used for app backgrounds to reduce visual fatigue and maintain a sense of calm.

- **Iconography:** Icons in the app are designed to be simple, intuitive, and emotionally neutral. They serve as visual anchors for navigation and interaction, especially for users with low digital literacy. Key icons (e.g., medication, sleep, calendar, journal, support team) are minimalist and rendered in blue or green depending on user role. Their consistent style and high-contrast design help users quickly recognize functions without cognitive overload. By avoiding overly playful or metaphorical visuals, the icons reinforce emotional clarity and usability.

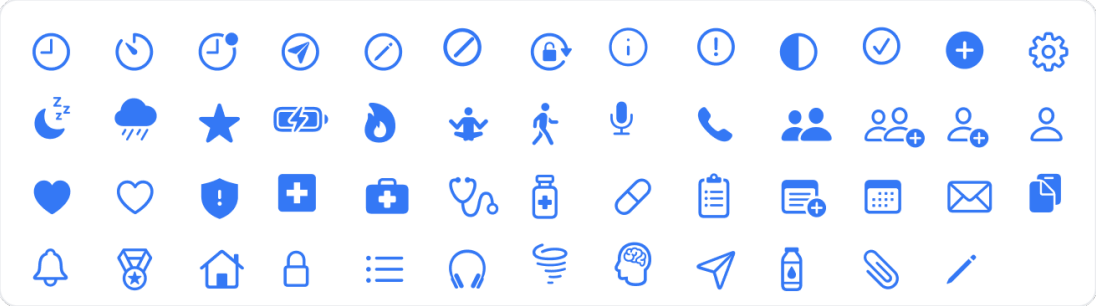


Figure 4.6: Iconography

- **Iconography and Apple's HIG** The iconography throughout the application adheres to Apple's Human Interface Guidelines, which recommend using simplified, universally recognizable shapes to reduce interpretation errors and cognitive load. Each icon whether representing mood, medication, messaging, or settings is designed in a minimalist line style to maintain visual consistency and align with the app's overall calming aesthetic. The use of monochromatic tones and consistent sizing ensures that icons support rather than distract from user tasks.

  This visual consistency is crucial in a mental health app, where user clarity

and emotional comfort are paramount. Moreover, the icons respect iOS system conventions, allowing users familiar with Apple products to navigate with confidence and ease. By grounding iconography in HIG standards, the app promotes intuitive interactions while maintaining accessibility and emotional safety.

- **Buttons and CTA Elements:** Each user role such as : bipolar patient, doctor, and support person interacts with the app through distinct, role specific buttons designed for clarity, accessibility, and visual consistency.

  Bipolar Patients interact with green buttons (#4DB6AC) that guide them through key actions such as submitting daily mood entries, saving personalized reports, or adding a healthcare provider. This primary color reinforces their active role in self-management.

  Doctors use blue buttons (#799FF0), which appear in the clinical dashboard. These buttons support navigational functions like "Next," "Save Notes," or "Add Patient," conveying professionalism and reinforcing trust in the clinical interface.

  Trusted Support Persons are presented with purple-toned buttons (#D0CAEB), symbolizing their secondary but meaningful contribution to care. These buttons are used for actions like sending supportive messages, reviewing shared reports, or setting reminders.

  All buttons are large, high-contrast, and consistently styled to support users with motor difficulties or attention challenges. Text labels are paired with intuitive icons, minimizing cognitive load and enhancing recognition. This structured, role-based use of color, shape, and function reinforces user expectations, reduces friction, and visually communicates each role's responsibility within the system.

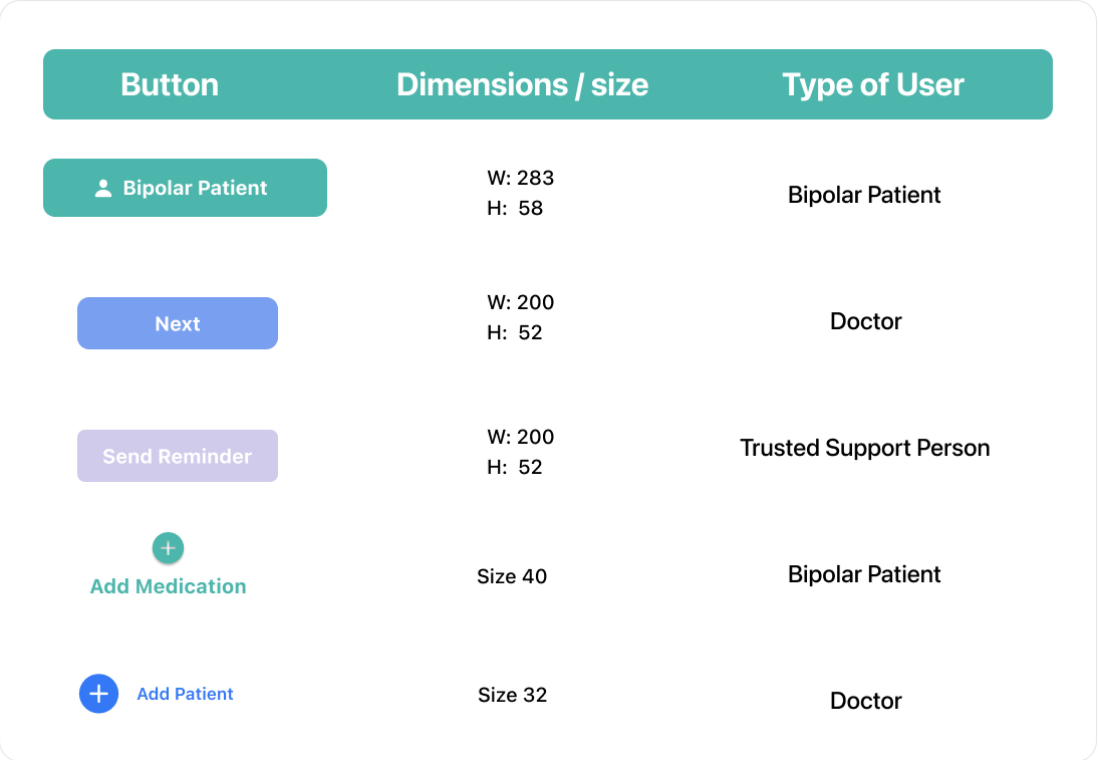


Figure 4.7: Button types, dimensions, and associated user roles within the application

4.2.2 UX Design Considerations

User experience decisions focused on reducing friction and enhancing emotional safety:

- **Navigation and Task Flow:** The application features distinct bottom tab bars for each user role of bipolar patients, doctors, and trusted support persons, ensuring that users are presented with only the most relevant tools and tasks according to their role. The bipolar patient dashboard includes a four-tab navigation bar: Mood Log, Calendar, Medication, and My Team. This setup supports quick access to core features related to daily tracking, adherence, and support, encouraging consistent engagement. The doctor's tab bar simplifies access to patient management and profile settings, focusing on clinical oversight and communication. In contrast, the support person dashboard offers a streamlined two tab layout "Today and

Calendar" designed to promote ease of use while still allowing meaningful interaction with the patient's emotional and medication data, based on permissions granted. These role-specific navigation flows reduce user confusion, improve usability, and support regular participation in care-related activities.

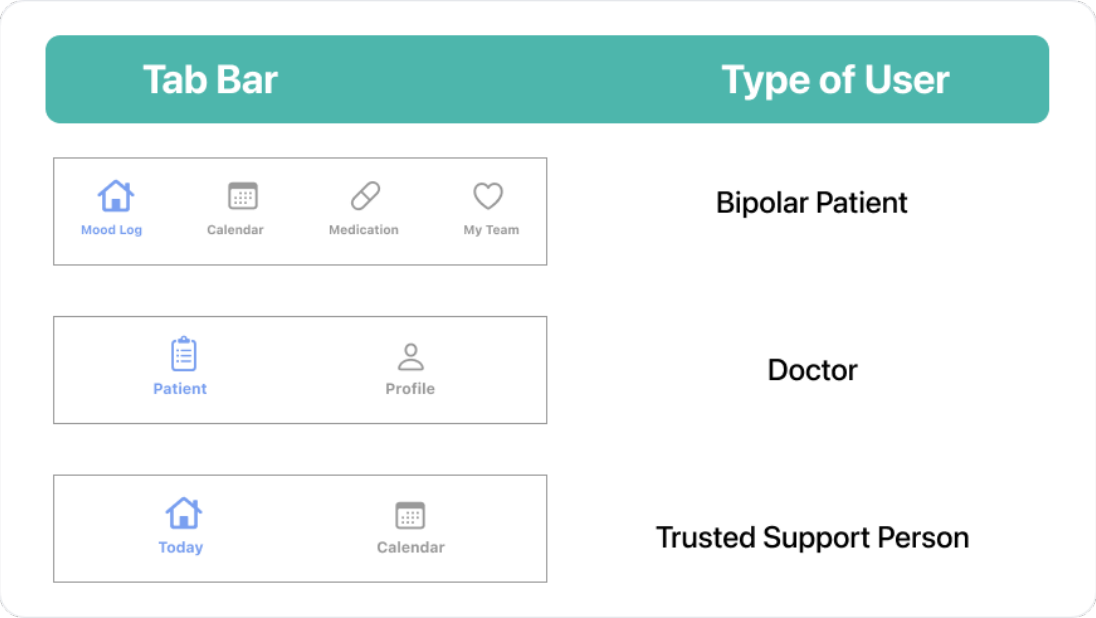


Figure 4.8: Role-Based Tab Bar Configurations

- **Multi-User Role Support:** The application includes three distinct dashboards designed respectively for patients, doctors, and trusted support persons each tailored to the unique responsibilities and access levels of the user type. The patient dashboard focuses on self-monitoring and emotional support, allowing users to record daily mood symptoms, receive AI-driven predictions, manage medication routines, engage with journaling features, and communicate with their care network. The doctor dashboard enables clinicians to oversee multiple patients by reviewing mood trends, monitoring medication adherence, accessing detailed reports, and managing appointments and direct communication. The trusted support person dashboard is designed for family members, partners, or friends, who can play a collaborative role in care based on permissions granted by the patient. These users can track mood and medication adherence, explore predictive mood

cycles, view reports, and send supportive messages, actively contributing to patient ongoing emotional regulation and treatment participation.

- **Emotional Reinforcement:** Gentle affirmations such as "You're doing great, just check in with yourself" aim to encourage use without judgment. Emotional safety is reinforced through goal-setting prompts, journaling features, and calming design choices.

- **Engagement Tools:** A point-based reward system incentivizes consistent use and builds healthy routines essential for bipolar disorder self-management [34].

- **Real-Time Feedback and Error Prevention:** Notifications for missed medication doses or incomplete entries support adherence and reduce disruptions in treatment plans [48].



Figure 4.9: Real-Time Medication Reminder Notification Supporting Adherence and User Engagement

4.3 Navigation Structure and Screen Flow, and User Dashboards

The navigation structure and screen flow of the application have been meticulously designed to prioritize clarity, accessibility, and ease of use, particularly for individuals managing bipolar disorder. Given the emotional and cognitive fluctuations associated with the condition, the user interface adopts a straightforward and supportive layout that simplifies task execution while reducing cognitive load.

Main Navigation The primary navigation is structured around a bottom tab bar that varies according to the user role—patient, doctor, or trusted support person. For patients, the tab bar provides access to Mood Log, Calendar, Medication, and My Team. For doctors, the interface offers streamlined access to patient data and communication tools without a tab bar. For trusted support persons, the tab bar includes access to mood overview and calendar views, focusing on monitoring and encouragement rather than full functionality.

Mood Log and Daily Entries The Mood Log is the core feature of the patient interface. Each day, users are prompted to complete a brief mood assessment consisting of seven sliding-scale questions addressing anxiety, sleep quality, energy, and overall emotional state. After completing the entry, users receive visual feedback via graphs and charts that help them understand their emotional trends over time.

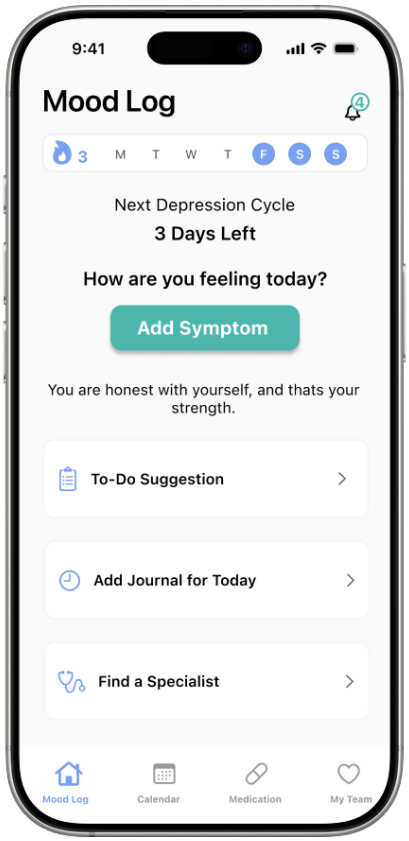


Figure 4.10: Mood Log and Daily Entries

Calendar and Visual Data The Calendar section allows users to view mood trends, missed doses, and predicted mood cycles across daily, weekly, and monthly views. It integrates mood data with medication adherence and symptom trends to support proactive management and facilitate informed self-care decisions.

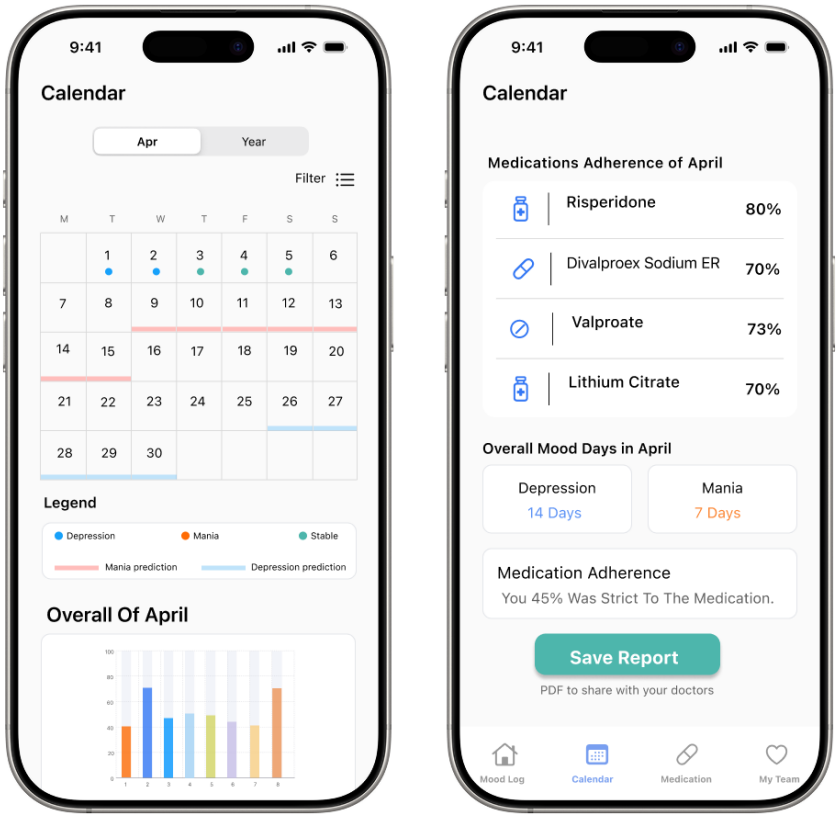


Figure 4.11: Calendar and Visual Data

Medication Tracking and Reminders The Medication section supports users in managing their prescribed treatments. It allows users to add medications, set up reminders, and log adherence through scheduled alerts and confirmation prompts. Users receive real-time notifications for missed doses or upcoming medications, promoting routine compliance and therapeutic consistency.

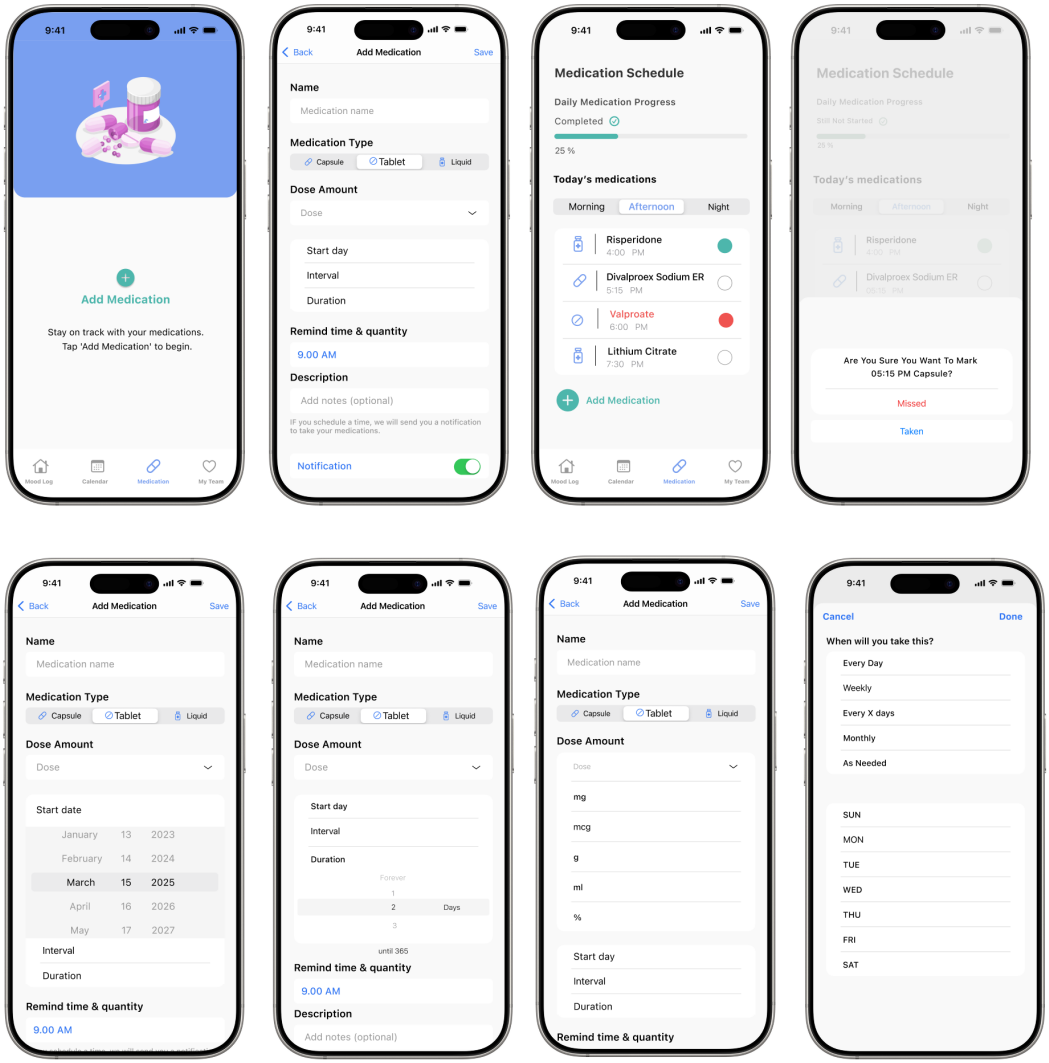


Figure 4.12: Medication Tracking

My Team – Support and Permission-Based Sharing The My Team section enables patients to build a care network by inviting trusted support persons (e.g., family, partners) and healthcare providers. Through an invitation code system, users can assign access permissions for mood tracking, medication adherence, or both. This permission-based sharing allows doctors and support persons to view relevant health data, while the patient retains full control over what is shared. Support persons can send motivational messages and track overall trends, helping build emotional and treatment continuity.

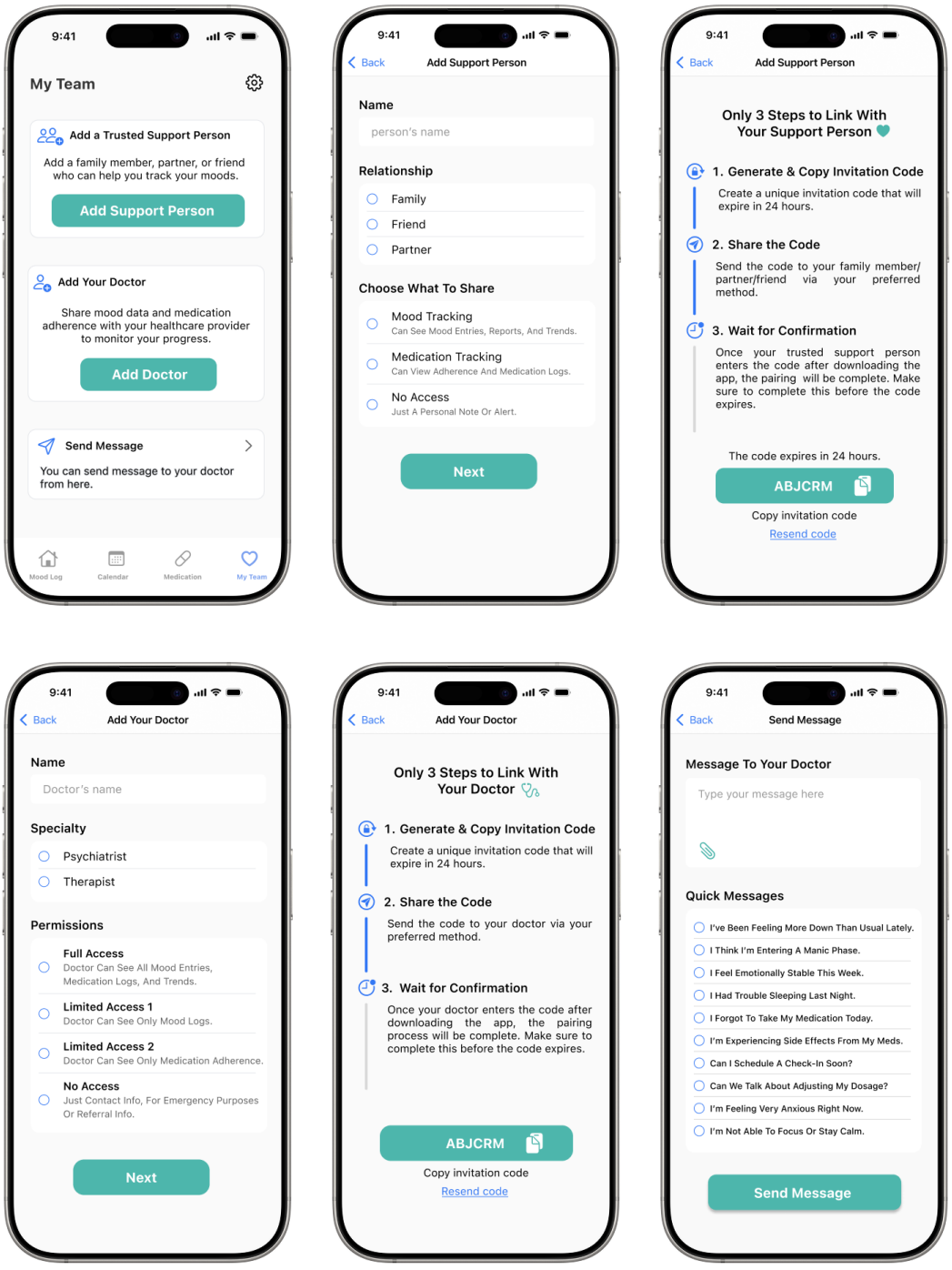


Figure 4.13: My Team "Support and Permission" Based Sharing

Screen Flow The app's screen flow is designed to be intuitive and sequential. Each task naturally leads to the next such as completing a mood entry, reviewing

data, and adjusting medication reminders allowing users to complete daily tasks with minimal friction. Emergency actions and settings are placed in accessible, predictable locations to minimize confusion and support emotional regulation during distressing periods.

4.3.1 Role-Based Dashboards and Care Collaboration

To support distinct user needs, the application includes three customized dashboards, each tailored to the user's role and responsibilities within the care ecosystem. The conceptualization of the three distinct user roles *patient*, *doctor*, and *trusted support person* was based on shared care models in mental health literature and benchmarking of existing digital health applications. While no direct user interviews were conducted, this role structure was informed by empirical studies emphasizing triadic collaboration for improved outcomes in bipolar disorder management [49, 50].

Each role's interface and permission level was designed to align with the typical responsibilities and interaction needs identified in clinical and digital health research. For example, patients have full access to mood tracking and personal data management; doctors view structured reports and adherence summaries; and trusted support persons receive limited, permission based updates to assist in daily care without infringing on patient autonomy.

Patient Dashboard Designed for simplicity and emotional comfort, this interface enables patients to log symptoms, view mood trends, track medication adherence, and access support tools. With access to all four core tabs *Mood Log*, *Calendar*, *Medication*, and *My Team*—the dashboard encourages self-awareness and proactive mood management. The layout and flow are optimized for daily usability, promoting user independence through motivational messaging, visual trends, and behavioral prompts.

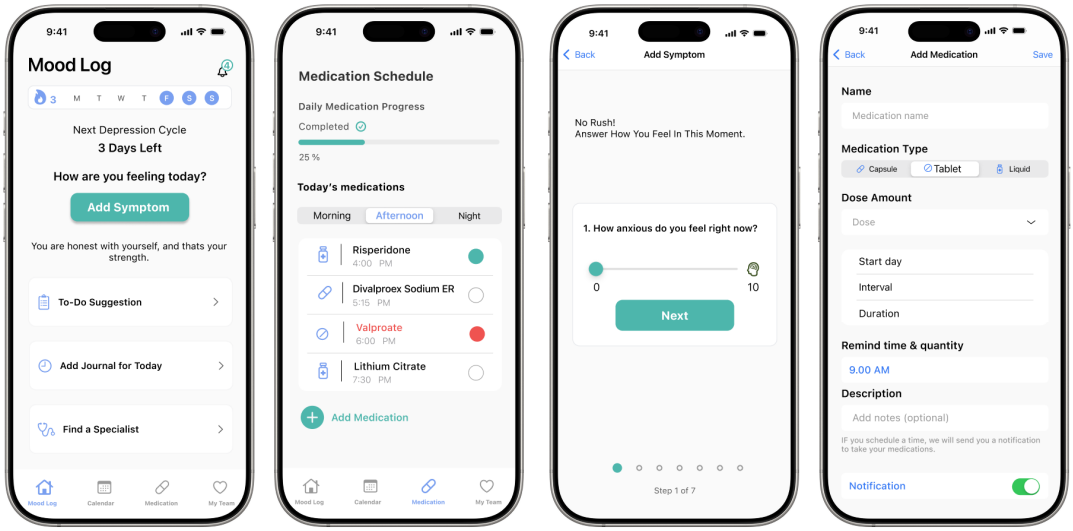


Figure 4.14: Patient Dashboard

Doctor Dashboard Built for efficiency and clinical clarity, this dashboard eliminates the tab bar in favor of a direct patient list interface. Doctors can access each patient's detailed summary, including emotional patterns, medication progress, and symptom reports. Features such as *Schedule Appointment*, *Send Message*, and *Add Note* support asynchronous care coordination and remote monitoring. The structured data view helps clinicians make timely, informed decisions and facilitates shared decision-making with patients.

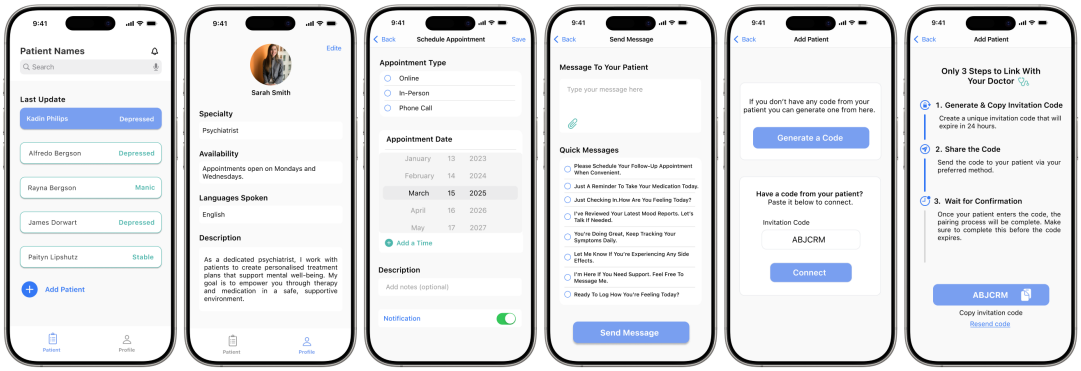


Figure 4.15: Doctor Dashboard

Trusted Support Person Dashboard With the patient's consent, family members, partners, or caregivers gain limited access to emotional and medication data via a simplified dashboard. The interface includes mood reports, medication adherence status, and a messaging tool for offering encouragement. Access levels are permission-based and can be adjusted by the patient. Trusted users cannot edit data but can submit support messages and request enhanced access. This role reinforces emotional safety and accountability, extending support beyond clinical settings.

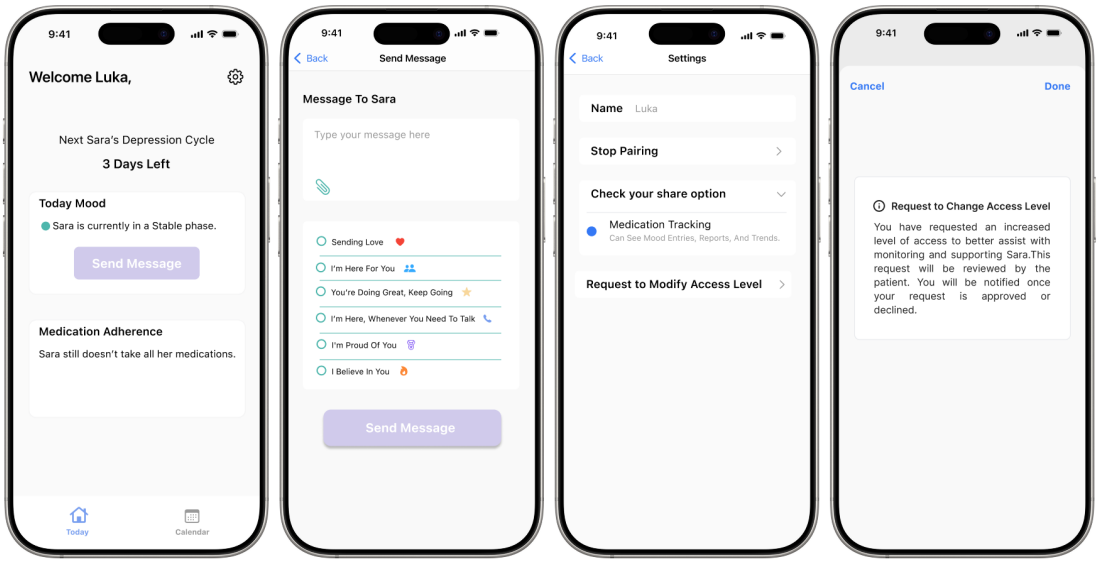


Figure 4.16: Trusted Support Person Dashboard

By offering customized workflows and navigational layouts across these three user roles, the application ensures that every stakeholder, from patient to caregiver to clinician, has an appropriate and empowering interface. This design supports the broader goal of enhancing collaboration, increasing adherence, and fostering emotional resilience in the ongoing management of bipolar disorder.

Consistency and Familiarity Across Dashboards Consistency across user roles patients, doctors, and trusted support persons was a critical part of the design system. While each dashboard uses distinct accent colors to visually differentiate user types (green for patients, blue for doctors, pink for support persons),

the structural layout, icon positions, and interaction logic remain the same. This consistency reduces learning curves and supports better accessibility for users who might switch roles or collaborate across roles.

Maintaining visual and interaction consistency aligns with Apple's HIG emphasis on user familiarity and minimal surprises. Users can rely on the same gesture patterns and navigation logic across all dashboards, ensuring that no additional cognitive load is introduced based on user type. This design strategy fosters a coherent and accessible multi-user experience.

4.4 Daily Mood Entry and Questionnaire System

The Daily Mood Entry and Questionnaire System is a core feature of the application, designed to facilitate daily tracking of mood states, behavioral symptoms, and overall well-being. The system comprises a concise, self-report questionnaire aimed at encouraging users to engage with the app daily to record their emotional and psychological experiences. This regular self-monitoring serves to capture real-time mood fluctuations and provide continuous data that can be reviewed by both the user and their healthcare providers.

The system includes seven key questions that focus on primary symptoms related to bipolar disorder, including anxiety, sleep disturbances, appetite, energy levels, and emotional states (e.g., feelings of hopelessness or happiness). Each question is answered on a scale from 0 to 10, allowing users to express the intensity of their emotions. This approach captures a comprehensive overview of mood-related symptoms while maintaining simplicity for daily use.

These questions were adapted from validated clinical tools such as the Internal State Scale (ISS) and the Mood Disorder Questionnaire (MDQ), ensuring that the app is based on well-established diagnostic frameworks suitable for real-world applications [2].

To encourage daily participation, the interface of the questionnaire is designed to be visually simple and intuitive. Each question is accompanied by positive,

affirming language that helps create a supportive environment, further promoting user engagement.

An important feature of this system is the visual representation of the mood data collected over time. The app generates charts based on the user's daily responses to the questionnaire. These visualizations allow users to track trends in their mood, identifying patterns such as periods of depression, mania, or stability. The charts display individual symptom levels and offer aggregated views of emotional trends over specific periods (e.g., daily, weekly, or monthly), helping users see their fluctuations in a clear, comprehensible format that motivates further self-monitoring and reflection.

For healthcare providers, the ability to view these visual reports aids in assessing treatment effectiveness, monitoring progress, and making informed adjustments to care strategies. Interactive graphs enhance collaborative care, empowering both patients and clinicians with actionable insights.

By integrating charts and data visualizations, the system fosters continuous engagement with the app, making it a valuable tool not only for emotional tracking but also for clinical monitoring. This data-driven approach ensures that users and their support systems can make informed decisions, track therapeutic progress, and better manage the challenges of living with bipolar disorder.

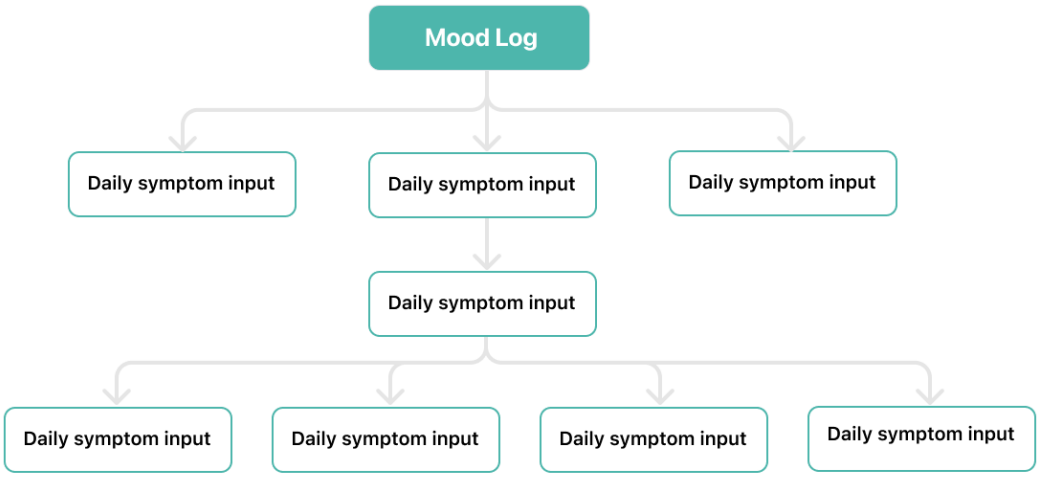


Figure 4.17: Data flow in the Predictive Mood Tracking System

This flowchart illustrates the transformation of user-entered data including daily mood ratings, sleep quality, and medication adherence into personalized predictions and support tools. The AI engine processes these inputs using a rule-based or learning-based model (e.g., decision tree or RNN) to forecast emotional states. These predictions are then visualized in the Calendar, paired with tailored To-Do Suggestions, motivational messages, and optional alerts for the user's care team. The goal is to support early intervention and strengthen both emotional regulation and collaborative care.

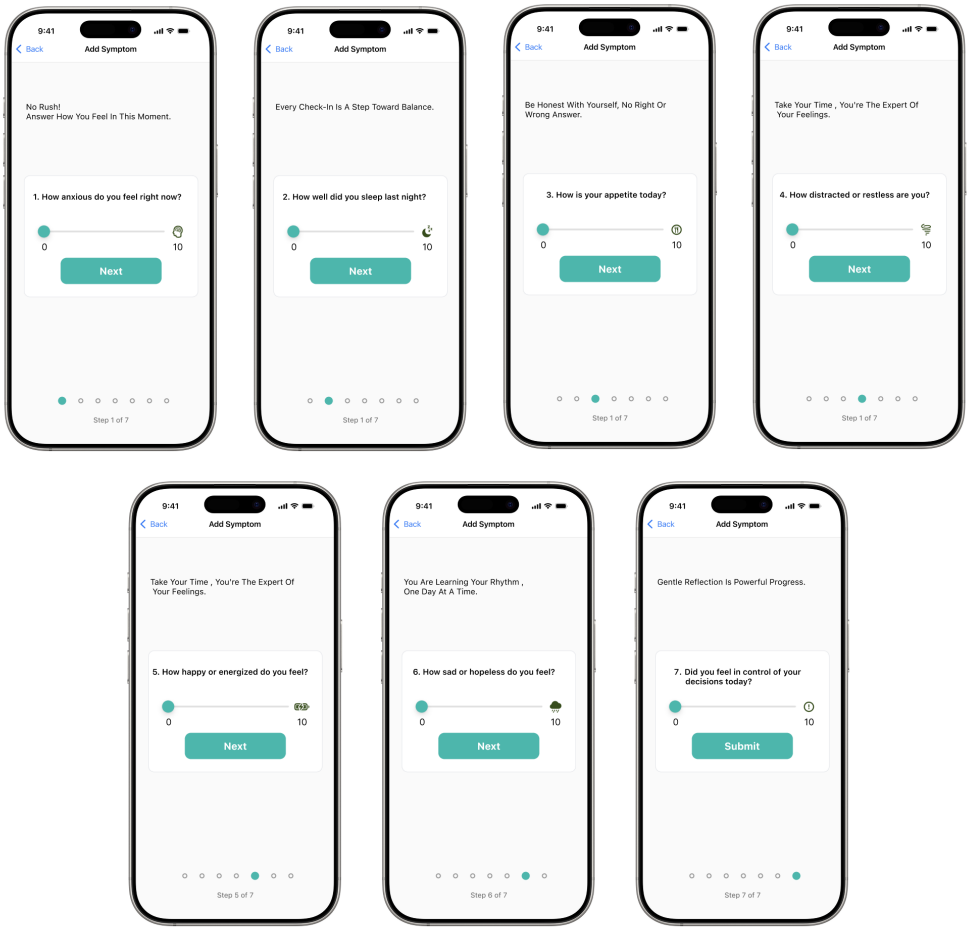


Figure 4.18: Daily Mood Entry and Questionnaire System

4.5 Visualization of Mood Trends

The **Visualization of Mood Trends** component plays a key role in empowering users, patients, doctors, and trusted support people, to track, interpret, and respond to emotional and behavioral patterns over time. By transforming complex mood and medication adherence data into intuitive graphical formats, the app supports emotional insight, facilitates early intervention, and enhances informed clinical decisions. Visual consistency is maintained across all dashboards, while slight stylistic variations, such as button color and labels, distinguish interfaces based on user roles.

Graphical Representation of Mood Data The calendar interface incorporates clear color-coded graphs to represent mood states - depression, mania, and stability - using blue, orange, and green, respectively. Each day is visually marked, and below the calendar, users can view daily bar charts that reflect symptom severity for factors such as sleep quality, anxiety, decision control, and medication adherence. These visual elements are designed for clarity, accessibility, and quick interpretation, ensuring that users with cognitive fatigue or low digital literacy can easily engage with the content.

The bar chart titled *"Symptoms of the Day"* uses a 0–10 scale for each symptom, providing users and clinicians with a snapshot of intensity per entry. Aggregating this data in one location avoids overwhelming users while supporting meaningful insight into emotional patterns.

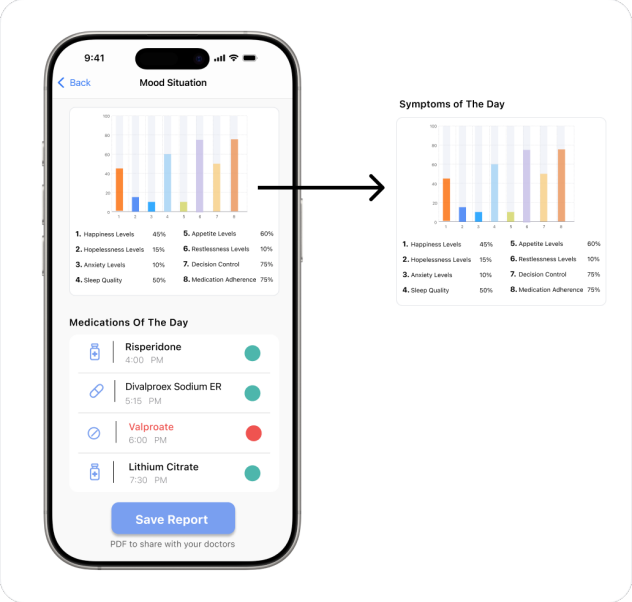


Figure 4.19: Symptoms of the Day Bar Chart

Trends and Predictions Beyond retrospective data, the app includes AI-powered predictions for future emotional states. The model analyzes mood patterns and predicts upcoming depressive or manic phases. These are displayed as subtle colored underlines in the calendar (red for mania, blue for depression), keeping users gently informed without causing anxiety. This predictive overlay encourages timely behavioral adjustments and supports proactive care by both

users and clinicians. Research supports that early recognition of mood shifts improves self-management and reduces symptom severity [27].

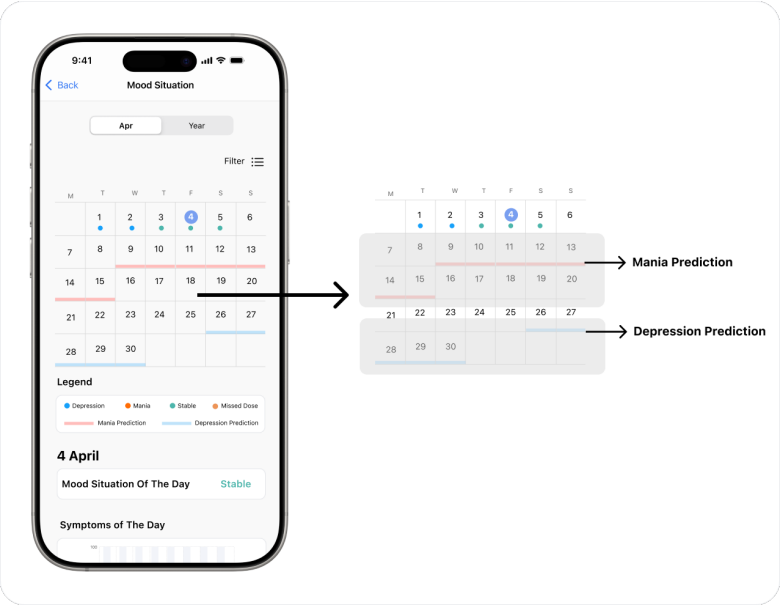


Figure 4.20: Calendar-Based Mood Episode Predictions

Integration of AI within UI The integration of AI-driven predictions is visually embedded in the user interface using subtle and non-intrusive design techniques. For instance, within the Calendar view, predicted manic and depressive episodes are indicated through soft color-coded underlines—orange for mania and blue for depression—beneath the date tiles. This gentle presentation ensures that users remain informed without being overwhelmed, preserving emotional safety.

Additionally, in the Mood Log interface, the AI system presents a brief, friendly sentence at the top of the screen, offering predictive emotional insight for the day. These predictive elements are visually de-emphasized but contextually relevant, in line with Apple's HIG principle of deference—allowing user-generated data to remain the focal point. By embedding AI functionality into the visual system in a calm and integrated manner, the app offers users continuous insights while maintaining clarity and emotional comfort.

Mood Correlation with External Factors The visualization system also integrates key behavioral data, such as medication adherence, sleep quality, and

lifestyle indicators, within the same interface. This allows users to recognize correlations, such as the impact of missed medication on emotional volatility or the link between poor sleep and depressive episodes. Such integration enhances emotional regulation through self-reflection and pattern recognition, aligned with findings that emphasize the importance of circadian stability and adherence in mood disorders [41, 43].

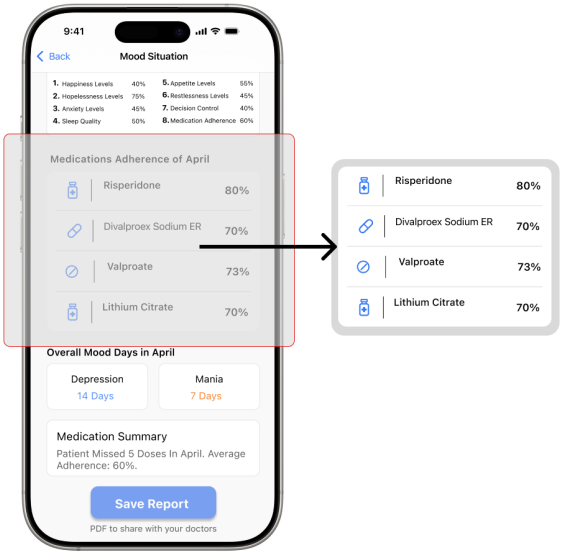
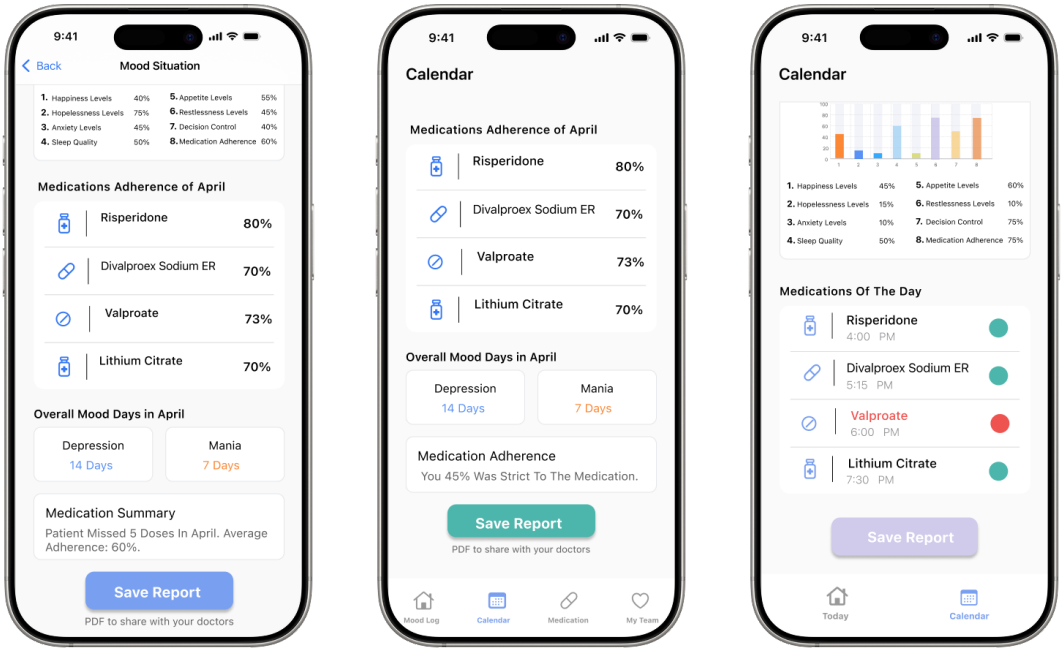


Figure 4.21: Visualizing Medication Adherence Alongside Mood Patterns

Data Sharing and Reporting Users can export their visual data into a professional PDF report summarizing mood trends, symptom ratings, and predictive insights. These reports are useful in clinical appointments or can be sent remotely, strengthening patient-provider communication. Doctors use them for longitudinal analysis and treatment planning, while support persons gain a better understanding of the patient's emotional state and can provide more targeted encouragement.



(a) Doctor Dashboard (b) Patient Dashboard (c) Support Dashboard

Figure 4.22: Calendar-Based PDF Reporting Across User Roles

Design Considerations In line with Apple's Human Interface Guidelines, the graphs use desaturated tones (e.g., green #4DB6AC, blue #799FF0, lavender #D0CAEB) and high-contrast text to ensure readability and avoid overstimulation. Graph legends are present on all charts, and spacing is optimized for vertical scrolling. Each dashboard retains the same structure to foster user familiarity, while the colors of the CTA buttons vary to maintain the identity of the role. This design approach supports clarity, consistency, and emotional comfort across the system.

4.6 Medication and Routine Management Interface

The medication and routine management interface plays a pivotal role in supporting users with bipolar disorder in maintaining consistent treatment adherence and monitoring their daily health activities. One of the core principles behind this interface is to empower users to track their medication schedules, ensuring they do not miss doses, which is crucial for managing mood disorders and avoiding

symptom exacerbation [37].

Medication Management System The medication management feature allows users to input their prescribed medications into the app and set reminders for daily or periodic dosing. Users can specify the type of medication (tablet, capsule, or liquid), dose amount, start dates, and frequency of medication (daily, weekly, monthly). A built-in reminder notifies the user when it is time to take each medication, after which they mark doses as "taken" or "missed," with visual feedback indicating adherence status. Such mechanisms are critical for fostering accountability and ensuring consistent adherence to prescribed medication [51].

Doctor Dashboard for Monitoring Adherence For healthcare providers, the app includes a dedicated dashboard that allows them to monitor patients' medication adherence and track missed doses. By reviewing the adherence calendar and associated data, doctors can intervene in a timely manner if a patient misses multiple doses or exhibits irregular medication behavior. This feature supports real-time communication between patients and their healthcare teams, aligning with shared-care best practices [42].

Encouragement through Reminders and Notifications To encourage routine adherence, the interface integrates persistent reminders and personalized motivational messages - such as gentle affirmations - to help users maintain a sense of accomplishment. Persistent notifications for missed doses or wellness tasks (e.g., taking a walk, drinking water) address the well-documented challenge of irregular adherence in bipolar disorder management [52].

Sharing Medication Data with Trusted Support Persons The app also allows users to link with trusted support persons (e.g., family members or caregivers), granting them limited access to medication adherence data. This collaborative feature strengthens the patient's support network, reflecting evidence that caregiver involvement can significantly improve treatment outcomes [53].

Data Export and Reports Finally, users can generate and export detailed adherence reports, including dose histories and missed dose summaries, to share with their healthcare providers. These reports facilitate transparent, data-driven discussions during consultations and enable clinicians to make informed adjustments to treatment plans.

4.7 App Architecture and System Flow

The App Architecture and System Flow are integral to the overall user experience, ensuring that the application provides seamless interaction for users with varying needs, such as patients, doctors and trusted support persons. From a design perspective, the system flow is structured to support not only functional ease of use but also emotional stability and user accessibility. This section elaborates on the system architecture, including the overall flow of the app, and highlights key design principles aimed at providing an emotionally supportive and user-friendly interface for individuals with bipolar disorder.

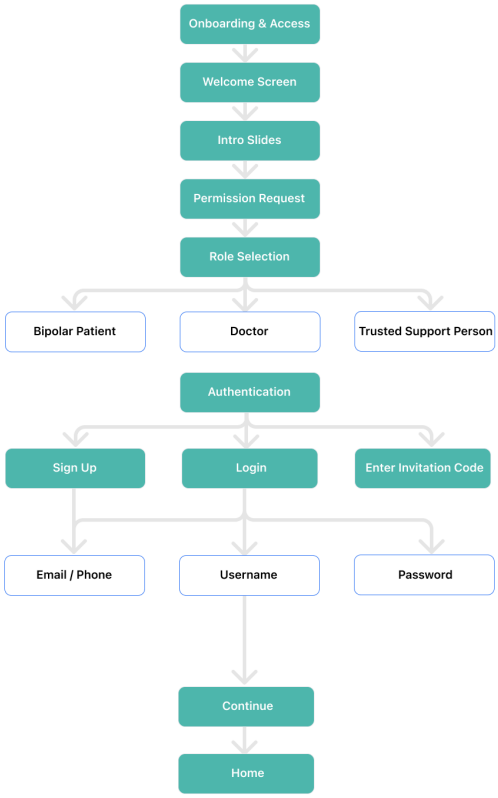


Figure 4.23: User On-boarding & Access Flow

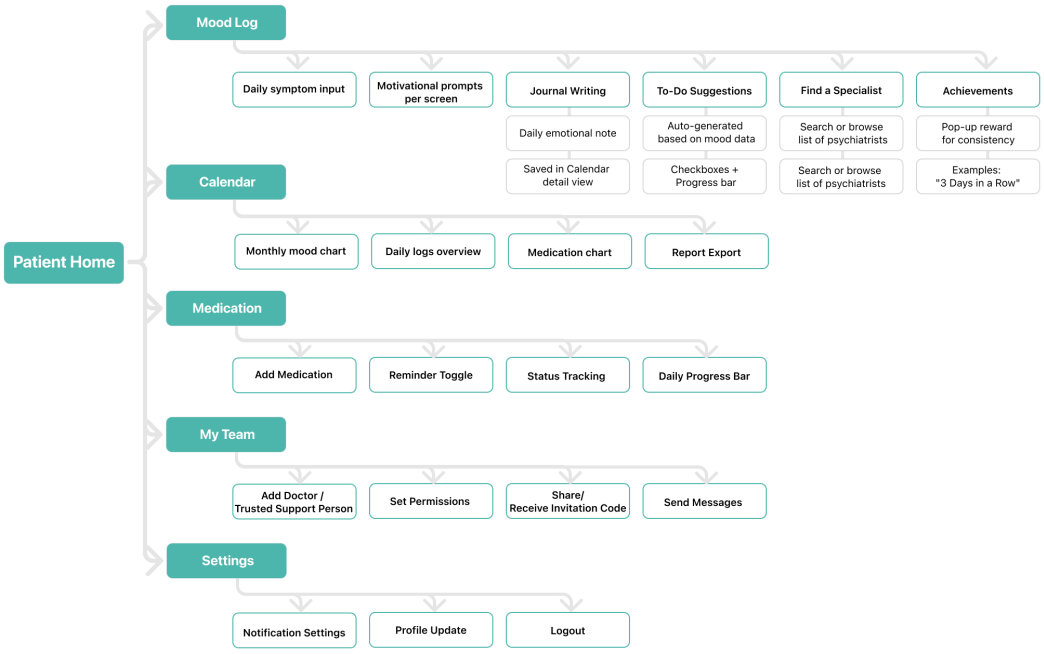


Figure 4.24: Patient Dashboard Architecture

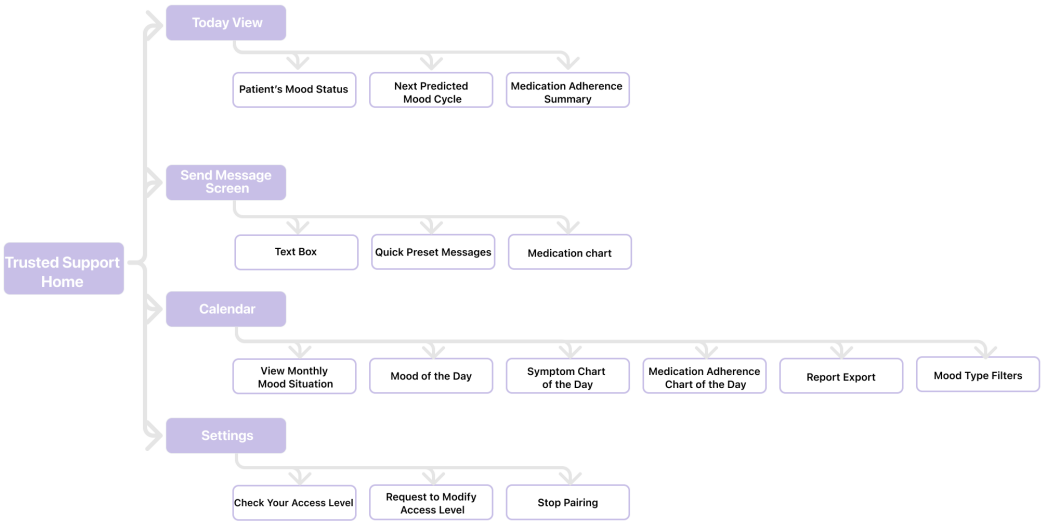


Figure 4.25: Trusted Support Person Dashboard Architecture

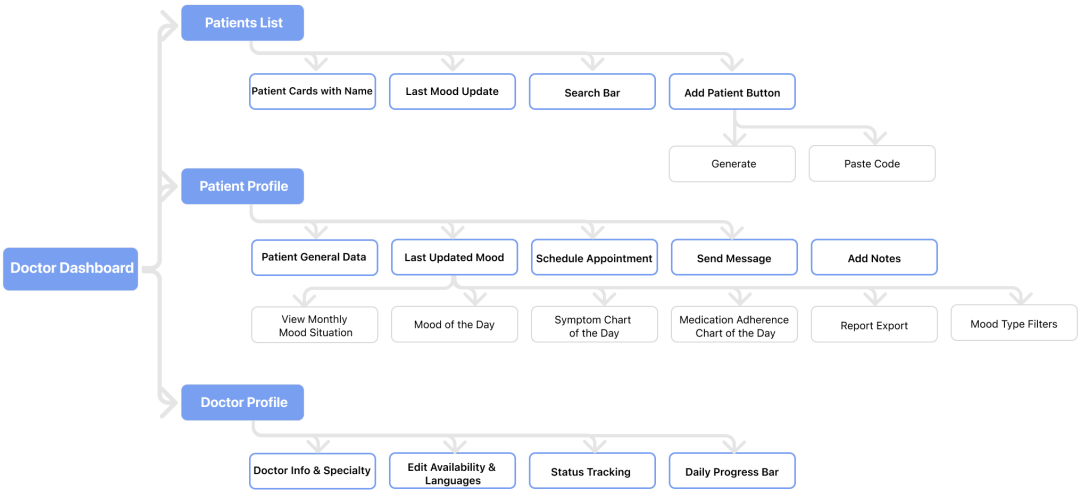


Figure 4.26: Doctor Dashboard Architecture

4.7.1 Conceptual System Architecture

From a design point of view, the system architecture is carefully designed to ensure that core functionalities, such as mood tracking, medication adherence, and communication with healthcare providers or trusted support persons, are easily accessible and intuitive to all users. While the technical aspects of data processing, storage, and security are managed by the backend systems, the design ensures

that users can efficiently navigate between essential features without unnecessary complexity.

Key design principles such as minimalist design have been applied to reduce cognitive load, which is particularly critical for users with bipolar disorder who may experience cognitive challenges, especially during mood episodes. The visual layout is intentionally simple, with an emphasis on clarity and intuitive navigation. For instance, the app allows users to effortlessly access mood tracking, medication adherence, and support tools, promoting ease of use and reducing the mental effort required for regular app interactions.

4.7.2 User Data Management and Privacy

Privacy and data management are central priorities in the design of the application. Given the sensitivity of mental health data, the app is designed to provide users with full control over their personal information. All mood-related and medication data is securely stored, and clear communication is made available to inform users about their data privacy rights.

The app features transparent data management controls, allowing users to make informed decisions about the sharing of their personal data. They can control whether or not their mood data or medication logs are shared with healthcare providers or trusted support persons. This functionality is carefully integrated into the app's interface, with simple options to enable or restrict access to different types of data. The design also facilitates obtaining user consent for sharing information with medical professionals, ensuring that the user's privacy is upheld while also providing flexibility in care management.

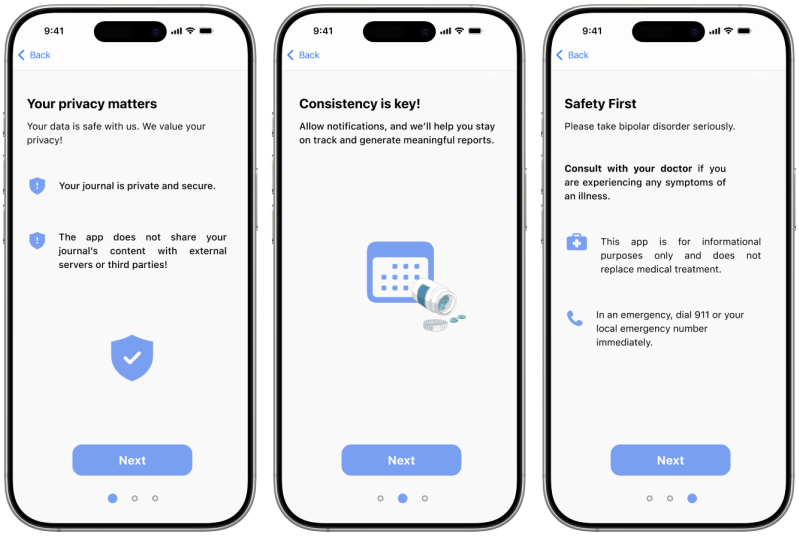


Figure 4.27: User Data Privacy, Safety, and Consent Awareness Screens

4.8 Integrating AI for Predictive Mood Tracking

The integration of Artificial Intelligence (AI) in the application enhances users' ability to predict mood fluctuations and proactively manage their bipolar disorder. By analyzing historical data from daily mood assessments, the AI system forecasts potential mood shifts, such as manic or depressive episodes. This feature allows users to anticipate emotional changes, take preventive actions, and reduce emotional distress. AI-driven mood prediction and personalized suggestions are embedded throughout the app, including in the Mood Log, Calendar, and To-Do Suggestions sections, helping users maintain better control over their emotional well-being.

4.8.1 AI for Predictive Mood Monitoring

The application utilizes AI-based predictive algorithms to analyze historical mood data collected through daily assessments. By identifying patterns in users' emotional fluctuations, the AI system forecasts upcoming mood states, alerting users

to potential episodes before they occur. Early detection and prediction of mood changes are essential for managing bipolar disorder, as discussed by Kannan, Smith, and Chen [25], whose research on AI-driven mood prediction models suggests enhanced forecasting accuracy and the potential for more proactive care. This predictive functionality is integrated into key sections of the app, providing users with early warnings and enabling them to take timely actions to stabilize their mood.

In the Mood Log section, users receive a brief daily reminder based on the AI's prediction of their emotional state for the day, such as an upcoming phase of mania or depression. This reminder not only prepares users for potential mood shifts but also empowers them to take steps toward self-regulation. By offering this early intervention tool, the app supports users in managing their emotional states more effectively, contributing to better emotional regulation and long-term well-being.

In the Calendar section, the AI predicts manic and depressive days using a color-coded system. Manic days are marked in orange, while depressive days are highlighted in blue. This color-coding allows users to visualize their predicted mood trajectory and take proactive measures. Fiorentino, Martínez, and Patel [26] highlights how predictive models in mental health applications help improve the accuracy of mood state classification and enable better preparation for mood shifts. By using this visual system, users can better understand their emotional patterns and plan ahead.

4.8.2 AI-Based Motivational Messaging and Activity Suggestion

In addition to predictive mood monitoring, the application also leverages AI to deliver personalized motivational support and behavioral suggestions, dynamically adapted to users' daily emotional states.

To complement the predictive mood-tracking functionalities, the proposed application integrates an AI-driven personalization system that dynamically gen-

erates motivational messages and daily activity suggestions based on users' self-reported emotional states. This approach is designed to improve emotional support, user participation, and adherence to self-care routines, critical factors for individuals who manage bipolar disorder.

The conceptual foundation for this feature is supported by recent advancements in AI-driven personalization models. For example, Signorelli, Rossi, and Chen [33] demonstrated that personalized motivational messages significantly improved user engagement and emotional resilience within the ADHERA CARING platform, a digital health tool originally developed to assist caregivers in pediatric healthcare settings. Although their target population differs, the principle of tailoring emotional support based on real-time emotional assessments is directly transferable to mental health management applications.

Similarly, Shetty, Wong, and Zhang [54] introduced the Self-Assessment of Emotions by Interactive Tool (SAEIT), which employs a two-dimensional valence–arousal emotional model to classify emotional states and deliver personalized feedback. By capturing daily emotional data across two simple axes: valence (positive to negative emotions) and arousal (calm to excited states), the system can suggest coping strategies, motivational prompts or behavior adjustments adapted to the user's immediate affective needs. [33, 54]

Building on these foundations, the app is designed to capture users' daily emotional input through a brief structured questionnaire. Based on the results, the AI system analyzes the emotional state and provides:

- A tailored motivational sentence that aligns with the user's current emotional condition;
- A recommended list of supportive activities, customized to help balance energy, regulate mood, or reinforce positive routines.

By integrating this dynamic personalization feature, the application not only predicts upcoming mood fluctuations but also proactively supports emotional well-being, fostering user autonomy, empowerment, and emotional self-regulation.

Mood State (User Input)	Motivational Sentence Example	Suggested Activity
Depressed, low energy	"Even small steps are victories today."	Gentle walk for 10 minutes
Anxious, high energy	"Pause, breathe, your calm is your strength."	5 minute deep breathing exercise
Neutral mood	"Consistency builds resilience."	Plan a simple self-care task

Table 4.1: Mapping of user-reported mood states to example motivational sentences and suggested activities

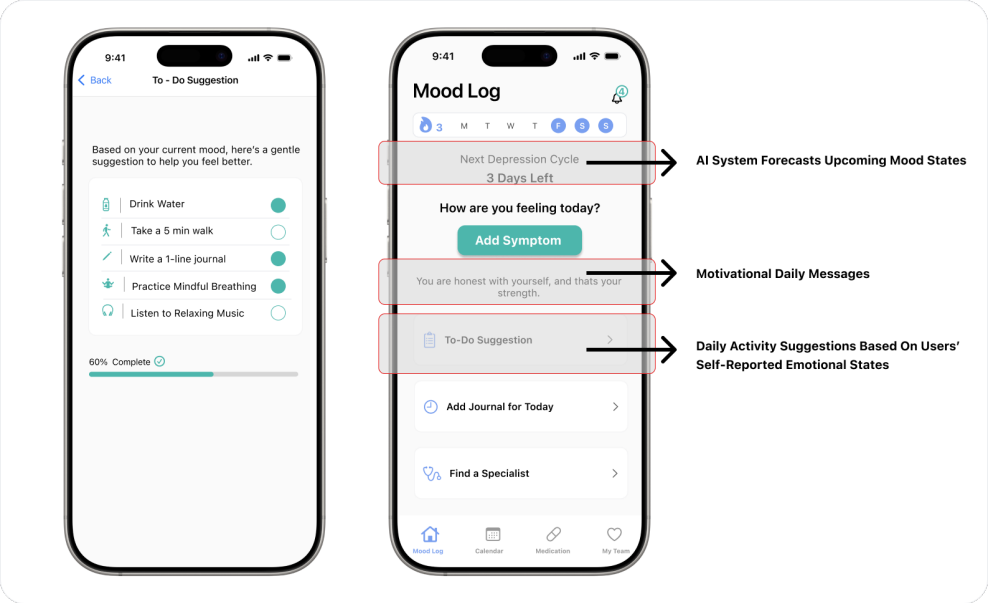


Figure 4.28: AI-Driven Mood Forecasting, Motivational Messaging, and To-Do Suggestions

4.8.3 Enhancing Emotional Self-Management

AI for predictive mood monitoring not only forecasts mood cycles but also empowers users by providing actionable suggestions for emotional management. The

combination of predictive analytics, motivational messaging, and task recommendations helps users better regulate their emotions and maintain consistent engagement with self-care practices. According to Shetty, Wong, and Zhang [54], interactive tools that offer personalized feedback based on emotional self-assessments can significantly improve user engagement and emotional well-being. The ability to track mood fluctuations and receive AI-generated suggestions further supports users in managing their bipolar disorder by enabling them to make informed, proactive decisions.

By integrating these AI-driven features, the app fosters a comprehensive approach to emotional self-management, allowing users to anticipate mood shifts, engage with personalized coping strategies, and maintain better control over their mental health. These tools contribute to enhanced emotional regulation, providing a supportive environment for users to stabilize their emotional well-being.

4.9 Challenges in Design and Implementation

Designing an application for individuals with bipolar disorder involves several challenges due to the unique nature of the condition, as well as the complexities of user needs, emotional support, and accessibility. In this section, we discuss the key obstacles encountered during the design and implementation phases of the app, focusing on issues related to accuracy in self-reported data, balancing simplicity with emotional depth, ethical considerations in mental health design, and user engagement.

4.9.1 Accuracy Challenges with Self-Reported Data

One of the primary challenges in the development of this app was ensuring the accuracy of self-reported mood data. Individuals with bipolar disorder may experience fluctuations in mood and behavior that can be difficult to articulate consistently. This inconsistency in self-reporting can affect the reliability of mood tracking, which is central to the app's functionality. To mitigate this challenge, the app integrates structured mood assessments, offering simple, intuitive questions that

reduce the cognitive load and increase the likelihood of accurate self-reporting. However, future improvements in data accuracy could include incorporating additional data sources, such as wearable devices, to passively track physiological markers that may correlate with mood changes.

4.9.2 Balancing Simplicity and Emotional Depth

Designing an interface that is both simple and emotionally supportive presented another challenge. It was essential to create an app that is easy to navigate and does not overwhelm users, particularly those who may struggle with cognitive overload during manic or depressive episodes. At the same time, the app needed to provide enough emotional depth to genuinely support users in managing their mental health. This was achieved by focusing on minimalist design principles and gentle language, ensuring that the app remained user-friendly without compromising on the emotional support necessary for individuals with bipolar disorder. However, striking the right balance between simplicity and emotional engagement remains an ongoing challenge, particularly as user feedback and real-world use cases evolve.

4.9.3 Ethical Considerations in Mental Health App Design

Mental health apps, especially those designed for users with bipolar disorder, raise several ethical considerations. One of the primary concerns is privacy and data security, as the app collects sensitive personal information about users' moods and behaviors. Ensuring the confidentiality of this data while still providing valuable insights and support to both users and healthcare providers is crucial. The app incorporates strong data encryption and offers users full control over their data, including permissions for sharing mood data with doctors or trusted support persons. Despite these measures, the ethical challenge of managing sensitive mental health data remains an ongoing concern, particularly as the app evolves and integrates more advanced technologies, such as AI and wearable sensors.

4.9.4 User Engagement and Long-Term Retention

Sustaining user engagement over time is another critical challenge. Research has shown that consistent self-monitoring and adherence to mental health tracking are essential for improving outcomes in individuals with bipolar disorder [35]. However, ensuring long-term engagement with the app can be difficult, as users may experience periods of instability where they are less motivated to engage with the app. To address this challenge, the app integrates gamification elements, such as a point-based reward system for completing daily mood assessments. These features encourage daily participation and help to build a consistent routine. However, ensuring that these features are both motivating and not overwhelming requires continuous user testing and feedback.

4.9.5 Design Inspiration from Apple Health App

Throughout the development process, Apple's Health app served as a key design reference. The app's use of structured information presentation, white space, and calm typographic hierarchy informed many of the decisions made in this project. In particular, the patient dashboard was intentionally modeled after the Apple Health app's "Senior Mode," where clarity, large fonts, and progressive disclosure support ease of use for vulnerable user groups.

This inspiration helped ensure the final interface would feel familiar to iOS users while introducing new functionality in a seamless and emotionally accessible manner. The structural simplicity and intentional design cues from Apple's Health ecosystem thus helped anchor this app's user experience in an already trusted digital environment.

4.9.6 Reflecting Beyond Visual Description: From What to Why

While the previous sections of this chapter have outlined the structural and visual elements of the application's design, it is equally important to consider the rationale behind these decisions. Rather than solely describing how features appear or

function, this section emphasizes the importance of articulating the underlying design motivations and their anticipated impact on user experience.

For example, the calendar interface was not only designed to display daily mood entries and symptom data but was deliberately structured to reduce cognitive overload by grouping visual summaries beneath each date. This layout supports users in drawing immediate associations between daily patterns and emotional states. Similarly, the choice to use soft, desaturated tones in the medication reminder system was guided by the goal of fostering a sense of emotional calm.

Additionally, several components were iteratively refined based on user feedback during early mockups. For instance, the placement and contrast of mood prediction indicators were adjusted to distinguish them more clearly from historical data points, addressing initial confusion observed in informal evaluations. These examples reflect an ongoing commitment to designing not only functional interfaces but also experiences that align with the emotional and cognitive needs of individuals with bipolar disorder.

By embedding such critical reflection within each design decision, the application goes beyond surface-level aesthetics to deliver purposeful and evidence-informed interactions.

4.9.7 Future Improvements

While the app addresses several critical design and implementation challenges, there are areas for further development. Personalization of the emotional support provided by the app, such as the integration of AI-based dynamic feedback, could be enhanced to better meet individual users' needs. Additionally, incorporating biometric data, such as from wearable devices, could improve the accuracy of mood predictions and provide users with more objective insights into their emotional states.

Chapter 5

Design Reflection, Contributions, and Future Work

This chapter presents a reflective summary of the app's conceptual development and anticipated impact, rather than empirical results. No clinical testing or user trials were conducted at this stage; therefore, the findings discussed are based on design analysis, literature review, and conceptual validation.

5.1 Key Takeaways from the Conceptual Design

The Predictive Mood Tracking App for bipolar disorder patients represents a significant advancement in the management of this complex condition. Through the integration of AI-driven mood prediction models, personalized motivational feedback, and collaborative care features, the app seeks to improve users' ability to manage their emotional well-being and prevent severe mood episodes. The design process, rooted in user-centered design principles, has resulted in an app that prioritizes both clinical relevance and emotional support, ensuring it meets the unique needs of individuals with bipolar disorder.

The key findings of the project highlight the following:

- Mood tracking is effectively integrated with AI-based predictive tools, providing users with early alerts about potential mood shifts, thereby enabling proactive interventions.

- The app's design incorporates personalized feedback and motivational messaging, which are tailored to users' emotional states, promoting self-regulation and adherence to treatment plans.
- Collaborative care features, which allow users to share their mood data with healthcare providers and trusted support persons, offer a holistic approach to managing bipolar disorder.

The app aims to address critical gaps in existing solutions, particularly the lack of predictive mood monitoring and personalized support for users. By integrating AI, emotionally supportive design, and real-time feedback, the app empowers users to actively engage in their mental health management.

5.2 Contribution to the Field of Digital Health and AI in Mental Health

While this thesis does not present results from clinical testing or direct user evaluations, its contributions are theoretical and design-based. These conceptual insights aim to inform future development in AI-assisted mental health tools, particularly those targeting bipolar disorder management.

This thesis contributes to the rapidly growing field of digital health by applying AI-driven predictive models to the management of bipolar disorder. The integration of AI for predicting mood shifts offers a novel solution to early intervention, allowing users to better anticipate mood episodes and take proactive steps to stabilize their emotional state. The incorporation of personalized feedback, in the form of motivational messages and activity suggestions, enhances user engagement, supporting emotional regulation and long-term adherence to treatment plans.

The proposed app concept aims to support the development of collaborative care models in mental health, where patients, healthcare providers, and trusted support persons can work together. Research has shown that such shared care models significantly improve treatment outcomes [42]. By enabling real-time

sharing of mood data, medication logs, and emotional insights, the app facilitates better communication between all stakeholders involved in the patient's care.

The design of the app introduces a personalized framework that may offer advancements beyond existing mood-tracking apps, which often lack predictive capabilities and contextual emotional support. This conceptual innovation is intended to support the daily management of bipolar disorder and contribute to improved mental health outcomes for users.

5.3 Future Directions and Potential Improvements

While the app offers a comprehensive solution for managing bipolar disorder, several areas remain for further development and enhancement:

- **User Testing and Feedback:** The app's design would benefit from real-world user testing to gather direct feedback from patients, healthcare providers, and support persons. This would allow for refinement of the app's user interface and user experience, particularly in the context of how the app is actually used in everyday life. Usability testing could further help in understanding how users interact with predictive features and personalized suggestions.

- **Integration with Biometric Data:** Incorporating data from wearable devices (e.g., Fitbit, Oura Ring) could improve the accuracy of mood predictions by providing more comprehensive and objective data on users' physical states (e.g., sleep patterns, physical activity, heart rate). This would enable a more holistic view of a user's mental and physical health, leading to better predictions and personalized suggestions.

- **Advanced AI Capabilities:** The app's AI predictive system could be further developed using deep learning techniques to enhance the accuracy and complexity of mood predictions. Future iterations of the app could also

incorporate ensemble learning models, which combine multiple algorithms to improve prediction accuracy [25].

- **Cultural Sensitivity:** To broaden its applicability, the app could be adapted to different cultural contexts to ensure that it resonates with a wider range of users. This includes adjusting the language, visuals, and personalized messaging to reflect cultural differences in emotional expression, self-care routines, and social support structures [17].

- **Expanding Collaborative Features:** Future versions of the app could include more advanced collaborative care features, such as real-time communication tools between patients and their healthcare providers or trusted support persons. This would allow for more dynamic, coordinated care and help ensure that users are fully supported in their journey toward mental health stability.

5.4 Final Thoughts

The Predictive Mood Tracking App represents a significant advancement in the management of bipolar disorder, offering an innovative AI-driven approach to predictive mood monitoring, personalized support, and collaborative care. By addressing critical gaps in existing mood-tracking applications—such as the lack of predictive capabilities and personalized emotional support—the app empowers users to take a more proactive role in their mental health management. It facilitates early detection of mood shifts, provides personalized feedback, and enhances treatment adherence, which are essential for effectively managing bipolar disorder. This innovative integration of AI helps users anticipate emotional fluctuations, reduce distress, and implement timely self-care strategies, contributing to improved emotional regulation and overall well-being.

The app's design marks a valuable contribution to the growing field of digital health for mood disorders. It provides users with personalized tools to track their moods, adhere to medications, and engage in emotionally supportive activi-

ties. The seamless integration of AI technology, clinical guidelines, and emotional support creates a comprehensive solution for individuals with bipolar disorder, ensuring they are better equipped to manage their condition effectively. By combining predictive mood monitoring and AI-driven emotional support, the app serves as a tool that not only manages symptoms but also promotes long-term mental health maintenance.

Although the app is currently in its design phase, its potential to improve user engagement, emotional regulation, and mental health outcomes is clear. The app has been structured to be scalable, with future integration of real-world user feedback and clinical validation being key factors in refining its functionality. As it moves into implementation and user testing, further adjustments can be made to enhance its usability, accuracy, and overall effectiveness in real-world settings. User feedback will provide insights into how the app is utilized in everyday scenarios, enabling improvements in areas such as usability, personalization, and predictive accuracy.

Moreover, biometric data integration and more advanced AI algorithms can significantly improve the precision of mood predictions and provide a holistic view of users' mental health. By integrating data from wearable devices (such as activity trackers, sleep monitors, etc.), the app can better capture physiological indicators of mood changes and refine the accuracy of its predictions. As the field of digital health evolves, further advancements in personalization through deep learning models will offer increasingly tailored solutions for individuals living with bipolar disorder.Future work should involve usability testing with real users and collaboration with clinicians to evaluate the practical effectiveness, user experience, and therapeutic value of the app design.

In conclusion, the Predictive Mood Tracking App holds immense promise as a transformative tool for managing bipolar disorder. By offering predictive mood tracking, personalized emotional support, and collaborative care, the app provides users with a comprehensive, proactive approach to mental health care. The work completed in this thesis serves as a strong foundation for further developments and future testing. With continuous refinement, user testing, and clinical validation,

this app has the potential to revolutionize how individuals with bipolar disorder manage their condition, leading to enhanced quality of life, better treatment adherence, and more effective long-term care.

5.5 Study Limitations

While this thesis presents a detailed conceptual design for an AI-driven mood tracking application, several limitations must be acknowledged. No real-world implementation of the application was developed, and the AI-based predictive system remains at the prototype stage without technical deployment or training on live user data. Similarly, the design has not been tested with actual users or clinicians, meaning there is no empirical data on usability, acceptance, or clinical effectiveness.

The feature requirements were derived from literature analysis and app benchmarking rather than direct stakeholder interviews or participatory design sessions. Furthermore, the proposed AI logic, such as decision trees and future RNN integration, remains conceptual and unvalidated in a real clinical environment.

These limitations reflect the project's early-stage, exploratory nature. Future work should include user testing, clinical collaboration, and iterative development to evaluate the app's real-world performance and therapeutic impact.

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